

Social Bee Lab

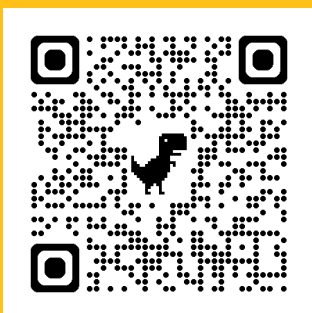
DISCUSSION PAPER SERIES

SBL-DP-00001

Publicly targeting group identities impacts take-up of educational opportunities

Manuel Munoz

December 2025



Publicly targeting group identities impacts take-up of educational opportunities

Manuel Munoz^{*} [†]

20.12.2025

Abstract

I investigate the unintended effects of publicly informing individuals that their selection for a beneficial opportunity, an international training program, is based on their group identity. In a natural field experiment with a Colombian university, I target 4,831 students and disclose to some that selection is based on demographics. Public identity disclosure reduced take-up by 27% and completion by 20% relative to a condition in which eligibility criteria were concealed. Three complementary online experiments with 1,113 students suggest that identity-based selection raises image concerns, which discourage program participation. The results highlight that the way institutions communicate beneficial opportunities can itself become a barrier to take-up.

Keywords: Diversity, Identity, Stereotype, Information disclosure, Image concerns

JEL Classification: C93, D03, D83, I21

^{*}New York University Abu Dhabi, Abu Dhabi, United Arab Emirates, and Universidad Autónoma de Bucaramanga, Bucaramanga, Colombia (e-mail: manumunoz@nyu.edu).

[†]I am grateful to Michelle Acampora, Carlos Alós-Ferrer, Joan Barceló, Rense Corten, Susan Dynarski, Moritz Janas, Mario Molina, Malte Reichelt, Ernesto Reuben, Pedro Rey-Biel, Sharlane Scheepers, Robert Stüber and participants at a number of seminars for helpful comments. I am also grateful to Jhon Alexis Díaz and Claudia Molina Gómez for valuable institutional support at Universidad Autónoma de Bucaramanga (UNAB), and to Alejandra Pérez for excellent research assistance. This study was approved by the ethics committee at Universidad Autonoma de Bucaramanga and pre-registered at AsPredicted #124501 (see Appendix H for details). I used OpenAI's ChatGPT for light copy-editing.

1 Introduction

Institutions and organizations increasingly design programs to support underrepresented or disadvantaged groups (e.g., STEM initiatives for women, up-skilling for immigrants, financial aid for low-income students).¹ To reach their intended audience, providers often explicitly highlight the targeted identities and the merits or needs that justify support.² While this practice signals an institutional commitment to inclusion, it relies on the assumption that making identity salient encourages participation.

Publicly informing individuals that they are targeted because of their group identities may, however, have unintended consequences. If individuals anticipate negative effects from an opportunity that is explicitly offered to them because of their demographics, public targeting may backfire. Rather than fostering inclusion and leveling the playing field, disclosing identity-based selection creates a behavioral barrier that reduces participation. Understanding these dynamics is critical: if the communication strategy alienates the intended beneficiaries, the policy risks undermining the very equity goals it aims to achieve.

In this paper, I report the results of a natural field experiment designed to evaluate how disclosing to individuals that they are chosen for an international training program because of their group identities impacts their choice to accept the offered opportunity.

I investigate this trade-off using a large-scale experiment conducted through a partnership between a university in Colombia and a top-tier American university. Through this alliance, I offered an international training program on non-cognitive skills to students who qualified based on specific demographic criteria (e.g., female, low socioeconomic status, or ethnic minority). A total of 4,831 eligible students received a personalized email inviting them to participate in the training program.

The content of the invitation varies between treatments, as the information about selection is either disclosed to the targeted individual and a third party (PUBLIC INFO), privately disclosed only to the target (PRIVATE INFO), or not disclosed at all (NO INFO).

¹ See [Alan and Ertac \(2018\)](#); [Alan et al. \(2019\)](#); [Carlana et al. \(2022\)](#). [Ko and Moffitt \(2022\)](#) provide a review of take-up for social benefits.

² [Dynarski et al. \(2021\)](#) examine high-achieving low-income students; [Carlana and La Ferrara \(2024\)](#) study underprivileged students with lower grades.

The main comparison of the field experiment is between PUBLIC INFO and NO INFO. For completeness, I also run the PRIVATE INFO condition, allowing me to separate the impact of private versus public disclosure.

I focus on two outcome measures: *take-up* is the primary outcome, and a target is said to take up the program when she completes the registration process after receiving the personalized invitation (i.e., the extensive margin). *Completion* is a secondary outcome, which occurs when a participant finishes all sessions of the program (i.e., the intensive margin). The main objective is to identify the behavioral consequences of public targeting for participation.

Among the 4,831 invited students, 1,407 (about 30% of the sample) took up the program and 1,066 (22%) completed it. The main finding of the study shows that take-up rates increase by 27% in the NO INFO condition relative to PUBLIC INFO. Publicly disclosing that selection is based on group identities significantly depresses participation rather than making students feel more recognized or included. This negative effect persists at the intensive margin, with completion rates rising by 20% when information on selection is not disclosed. The pattern is similar across high- and low-GPA students and across identity profiles, indicating that the detrimental effect of public targeting is robust.

A potential mechanism is that public targeting by group identities triggers image concerns (see e.g., [Bursztyn and Jensen 2017](#); [Leibbrandt and List 2018](#); [Moffitt 1983](#)). If accepting the invitation risks signaling pity, reinforcing stereotypes, or exposing individuals to judgment, some may forgo the opportunity even when it is personally valuable. To investigate this channel, I conducted three lab-in-the-field experiments with 1,113 students from the same university who were not invited to the training program. In each experiment, I vary whether participants are informed that selection is identity-based or not. I find evidence that the behavioral response is driven by image concerns: disclosing identity-based eligibility increases anticipated feelings of pity, stereotype threat, and social judgment. Consequently, a significant share of individuals forgoes a valuable opportunity to avoid the image costs associated with the targeting.

Finally, I quantify the costs of this behavioral friction by estimating the impact of the training on student outcomes. Using administrative panel data and a difference-in-differences design, I estimate that completing the program causally increases GPA by

0.034 standard deviations, with effects growing to 0.058 in later semesters. The reduction in take-up induced by the public disclosure of eligibility thus represents a tangible welfare loss. By discouraging participation, the targeting strategy inhibits human capital accumulation among the very groups the program aims to support.

The results of my work contribute to a prominent research agenda that studies why take-up rates remain low even when opportunities are clearly advantageous (for a recent review see [Bearson and Sunstein 2025](#)). This line of inquiry lies at the intersection of academic research and public policy, given the substantial investments by governments and private institutions in socially beneficial programs that are often underutilized because of low participation. Existing work shows that, beyond structural barriers such as limited time or resources, a range of behavioral frictions can depress take-up. These include difficulties in processing complex information (see e.g., [Bhargava and Manoli 2015](#); [Finkelstein and Notowidigdo 2019](#)), aversion to uncertainty about eligibility or benefits (see e.g., [Dynarski et al. 2021](#); [Burland et al. 2023](#)), and psychological costs associated with accepting potentially stigmatizing offers (see e.g., [Butera et al. 2022](#); [Moffitt 1983](#)). My study contributes to this agenda by showing that the way in which identity-based eligibility is communicated can itself become a barrier to participation, as it triggers image concerns.

A growing literature shows that image concerns shape individuals' choices when their actions are observable. For instance, [Bursztyn and Jensen \(2015\)](#) demonstrate that minority students avoid academically beneficial behaviors when these may signal an undesirable social type to their peers. [Bursztyn et al. \(2017\)](#) show that single women strategically reduce their expressed labor-market ambition when observed by classmates, consistent with marriage-market image concerns. In related work, [Bursztyn et al. \(2020\)](#) and [DellaVigna et al. \(2012\)](#) find that people incur real costs to avoid situations in which their behavior could be publicly revealed, and [DellaVigna et al. \(2017\)](#) provide evidence that social-image motives influence decisions such as voter turnout.

My study extends this work to a distinct and policy-relevant setting: the design of identity-based targeting in educational programs, as commonly used in positive discrimination and affirmative action initiatives (see e.g., [Bertrand et al. 2010](#)). Rather than focusing on eligibility rules, I isolate the behavioral costs of making group-based criteria salient at the individual level, showing that well-intentioned outreach practices

can undermine the equity goals they aim to advance. This mechanism helps explain why targeted opportunities may be underutilized, even when they are individually attractive, free, and institutionally supported.

These findings have direct policy implications for the design of diversity and inclusion initiatives. While emphasizing identity can signal an institutional commitment to equity, my results show that making identity salient in personalized invitations imposes a *behavioral tax* that deters a meaningful share of intended beneficiaries. A practical alternative is to determine eligibility using administrative data while omitting explicit references to group identity in the invitation. This is the NO INFO strategy in my field experiment, which preserves distributional goals by ensuring resources reach the intended groups, yet it shields individuals from the image costs that suppress take-up. By shifting the burden of identification from the individual to program providers, policymakers can reconcile equity objectives with efficiency (see e.g., Fryer et al. 2008; Finkelstein and Notowidigdo 2019; Dynarski et al. 2021).³

The rest of the paper is organized as follows. Section 2 describes the setting and experimental design. Section 3 presents the main results of the field experiment. Section 4 reports the complementary experiments on mechanisms. Section 5 concludes.

2 Experimental Design

2.1 Institutional Setting

This project is the result of a collaboration between an American global university (the international partner) and a university from Colombia (the local partner). The local partner is a private non-selective university with about 6,500 students. Evaluating identity-based targeting in this setting is ideal because, unlike public universities (which serve mostly low-income students) or private selective universities (which serve mostly high-income students), private non-selective institutions serve students from diverse social backgrounds (see Cárdenas et al. 2021; Londoño Vélez 2022). As part of the collaboration, the international partner agreed to support students who belong

³ Finkelstein and Notowidigdo (2019) and Dynarski et al. (2021) similarly use administrative data to identify eligible populations without requiring individuals to self-identify.

to at least one of several groups identified by the local university, including women, low socio-economic status (SES) students,⁴ ethnic minorities, rural-origin students, and first-generation college students. Students in any of these categories were directly offered the opportunity to participate in an international training program, described in Section 2.2.

Using administrative data, I identified 4,831 students who belonged to at least one of the targeted social categories. The share of these categories in the student population is as follows: female (55%), middle social class (45%), low social class (31%), rural (24%), first generation (15%), and ethnic minority (3%), where about 76% of students held at least one category. After the experiment concluded, the program was offered to all remaining students who had not been invited earlier, ensuring that no student was excluded from the opportunity on the basis of demographics.

2.2 The Training Program

The training program is offered as a selective opportunity and participation is by invitation only. The content of the program is centered around *self-efficacy* and the development of non-cognitive skills to set and achieve goals (see Milkman 2021). The topic of the program was curated so that it could be of interest and benefit to participants irrespective of their major, year of study, and other relevant characteristics.⁵

To motivate take-up and reduce participation costs, the invitation is personalized and explicitly states that the student already has a guaranteed slot in the program, eliminating uncertainty about eligibility and access. The program is free of charge and consists of nine pre-recorded, 30-minute online sessions. Crucially, the training was asynchronous and completed individually. This design ensures that participation decisions are not confounded by concerns about peer composition or forced social interaction

⁴ Colombian households are classified into six strata based on dwelling characteristics, a system determining utility subsidies and serving as a standard income proxy (Bogliacino et al. 2018). I use *low socio-economic status* (SES) to refer to the lower strata (locally termed *clase media baja*).

⁵ Unlike specialized skills such as coding (see, e.g., Carlana and Fort 2022), this topic was selected to be relevant across all majors and social backgrounds.

with other targeted students.⁶

The full schedule was provided at the outset, with two sessions launched each week (on Mondays and Thursdays), enabling students to track their progress and plan flexibly at their own pace. The program had no prerequisites and was open for participation irrespective of which courses students had taken so far.⁷ Access constraints were minimal: the university provides multiple computer rooms and free campus wifi, so students could participate even without personal devices or internet at home.⁸

Beyond the course content, participants received a completion certificate indicating that the program was taught by faculty from an internationally recognized university, a credential that can strengthen applications for jobs, internships, or scholarships (see e.g., [Athey and Palikot 2024](#)). The certificate intentionally omits any mention of the targeted social categories. This prevents concerns that listing the credential on students' CVs might inadvertently send a negative message to employers.⁹ Participants who completed the program also entered a lottery for two recent-generation iPads.

By combining low participation costs with both symbolic and material benefits, the design aims to minimize standard structural and behavioral barriers to take-up, allowing a cleaner test of the effect of public targeting.

2.3 Invitation Messages

Each selected student received a personalized invitation email from the local university's Office of International Relations. Because this office routinely advertises international opportunities, the email resembled a standard institutional communication rather than

⁶ Peer composition concerns can deter educational investment (see e.g., [Bursztyn and Jensen 2015](#); [Bursztyn et al. 2019](#)). To isolate the effect of targeting from such social concerns, the program was designed as an asynchronous, individual training similar to a Massive Open Online Course (MOOC), with no participant interaction.

⁷ The design of the program draws on several behavioral insights: (i) certainty about eligibility increases participation ([Tversky and Kahneman 1986](#)); (ii) a zero price increases valuation relative to low positive prices ([Shampanier et al. 2007](#); [Burland et al. 2023](#)); and (iii) combining routine scheduling with flexibility facilitates completion ([Beshears et al. 2016, 2021](#)).

⁸ By the 2022 launch all COVID-19 restrictions had been lifted, ensuring full access to campus facilities.

⁹ Explicitly signaling membership to stereotyped groups can negatively impact labor market outcomes ([Bertrand and Duflo 2017](#)).

a research invitation.¹⁰ All messages were sent to students' institutional email accounts during the academic term, when these accounts are actively used for course communications, maximizing the likelihood that targeted students would see the invitation.

The email informed targeted students about the collaboration between the local and international universities and explained that, as part of this agreement, the international partner was offering a training program to help them acquire or further develop skills to set and achieve goals. It outlined the program content, participation benefits, and *selection criteria*. I experimentally varied how this selection information was disclosed to manipulate the way individuals were targeted. In the invitation email (see Appendix A), a randomly chosen subset of students received the following message:

You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA].

The remaining students received a longer version that explicitly stated that selection was based on group identities:

You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA], and because you fulfill one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a first-generation student (neither of your parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

Students who received this longer message learned that their group identities influenced their selection, whereas for those who received the shorter message, this role of their identities was not disclosed.¹¹

Across treatments, the invitation email also informed students that, to register for the program, they needed to ask a faculty member to send an endorsement email on their

¹⁰ The program was offered through the university's internal channels, and students received the invitation as part of the standard flow of institutional opportunities, which classifies the intervention as a natural field experiment (see Harrison and List 2004).

¹¹ The wording mirrors standard equal opportunity statements by listing targeted groups jointly (see, e.g., Leibbrandt and List 2018). While this framing could theoretically trigger aversion to associating with other targeted groups, the program's strictly individual design mitigates this concern (see Section 2.2).

behalf to the program’s account. I designed this step to test social image concerns by bringing a third party into the targeting process. Each student received a pre-defined endorsement template whose content varied experimentally but mirrored the information provided in the main invitation. For example, in the baseline version the endorser was asked to write:

I, [Professor’s name] endorse student [Student’s name] to take part in the training program..., because he/she can benefit from this program, as his/her cumulative GPA is [Student’s GPA].

For a randomly chosen subset of students who received the longer invitation, the endorsement template additionally made selection based on group identities explicit:

I, [Professor’s name] endorse student [Student’s name] to take part in the training program... because he/she can benefit from this program, as his/her cumulative GPA is [Student’s GPA], and because he/she fulfills at least one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a first-generation student (neither of his/her parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

All students needed an endorsement to register for the program, but only a subset used an endorsement template that revealed to the third-party endorser that their demographic characteristics were part of the selection criteria.

2.4 Treatments

I designed three experimental treatments that vary whether the eligibility criteria are disclosed to both the targeted individual and a third party, to the target only, or to neither. The aim is to test how these variations in information disclosure impact take-up and completion rates.¹² I conducted the field experiment in two separate waves, each

¹² While *take-up* typically denotes receiving a benefit for which one is eligible (Bearson and Sunstein 2025), I define it here as the decision to register (the extensive margin). *Completion* is defined as finishing all nine sessions (the intensive margin), as the certification and lottery incentives were conditional on full participation.

targeting students with different levels of *academic performance*. Table 1 summarizes the features of each treatment as well as the number of individuals invited in each wave.

Table 1 Experimental Treatments and Design

The top panel details the disclosure of selection criteria to the targeted student and the faculty endorser for each treatment condition. The bottom panel reports the number of invited students (N) assigned to each condition in Wave 1 (High Performance) and Wave 2 (Low Performance).

	Treatments		
	PUBLIC INFO	PRIVATE INFO	NO INFO
Information is disclosed			
<i>To student</i>	✓	✓	×
<i>To endorser</i>	✓	×	×
Invitations per wave			
<i>High performance</i>	n=864	n=864	n=833
<i>Low performance</i>	n=776	n=757	n=737

Next, I describe the treatment variations, which were implemented identically across both waves of the study.

PUBLIC INFO: Targeted students are explicitly informed that they are invited because of their demographic characteristics, and the third-party endorsers receive an endorsement template stating that selection is based on these group identities.

PRIVATE INFO: Targeted students receive the same identity-based explanation of their eligibility as in PUBLIC INFO, but the endorsement template sent to faculty does not mention group identities.

NO INFO: Targeted students are selected using the same demographic criteria as in the other treatments, but neither students nor endorsers are told that selection is based on group identities.

2.5 Sample and Implementation

The experimental sample consists of all undergraduate students at the local university who, based on administrative records, belonged to at least one of the targeted social categories described in Section 2.1: female, low socio-economic status, rural origin, first-generation college student, or ethnic minority. In total, 4,831 eligible students received an invitation to participate in the program during the 2022–2023 academic year. To accommodate institutional requirements, the field experiment was implemented in

two waves within the same academic year. Wave 1 (Fall 2022) targeted high-performing students ($N = 2,561$), and Wave 2 (Spring 2023) targeted low-performing students ($N = 2,270$). Within each wave, I randomly assigned students to one of the three information conditions (PUBLIC INFO, PRIVATE INFO, NO INFO) using block randomization based on the targeted categories. This procedure yields balanced samples across treatments in both waves (see Table A-1 in Appendix A).

A notable feature of the invitations, as part of the requirements from the partner university, is that students' GPA was mentioned explicitly. This allows me to study public targeting among both high-GPA students, who may view the invitation as a reward for merit, and low-GPA students, who may see it as remedial support. However, it also limits comparability across waves, so I refrain from directly comparing high- and low-GPA students and focus instead on treatment differences within each wave.

The core elements of the intervention, such as invitation scripts, randomization blocks, and outcome definitions, were fixed at the outset of the collaboration and implemented identically in both waves. The study was formally pre-registered at AsPredicted (#124501) prior to the launch of Wave 2, once the data-sharing agreement for scientific publication had been finalized. The timing of the pre-registration reflects these administrative requirements rather than any change in design: no additional outcome variables were introduced after completing Wave 1, and no data were collected for other research purposes (See Appendix H).

Invited students had two weeks from the date of the email to register for the program (take-up). After the registration window closed, two sessions were launched each week, and registered students had five weeks to complete all sessions. I define completion as finishing the nine sessions of the training program.

Because treatment was assigned at the individual level, a potential concern is that students might discuss the invitation with peers in different conditions, generating spillovers. To test for this, I proxy peer exposure using administrative course-enrollment records, where two students are defined as *peers* if they shared classes. Unlike residential universities where housing drives social networks (see, e.g., [Sacerdote 2001](#)), this institution is non-residential; consequently, shared classes constitute the primary mechanism for peer interaction and information diffusion. I estimate whether a student's take-up decision is influenced by the share of their peers invited to the training

program. Appendix C reports the network analysis and shows no evidence of peer treatment spillovers. The treatment effects remain stable regardless of the number of treated peers in a student’s network, suggesting that the mechanism is a direct private response to how they are targeted rather than a socially diffused information effect.

2.6 Hypotheses

To derive testable hypotheses, I adapt the framework of Bursztyn and Jensen (2017) to my setting, focusing on how public targeting affects the decision to participate in the training program. In this random utility model, an individual takes the program if the (unobserved) net utility of participation is positive. Specifically, there is a latent variable $\tilde{a}_i$ capturing the desirability of participating in the program, so that $a_i = 1$ (i.e., i takes up the program) if $\tilde{a}_i \geq 0$ and 0 (no take-up) otherwise. This is expressed in the following utility function:

$$\tilde{a}_i = B - C + \phi_i + \epsilon_i \quad (1)$$

where B is the direct benefit from the program, C is the cost of participation, ϕ_i captures the identity-related consequences of targeting, and ϵ_i is an idiosyncratic shock. I initially treat B and C as constant across participants and focus on the role of ϕ_i (I later relax this assumption).

If disclosing that selection is contingent on group identities helps individuals feel seen and valued, then $\phi_i > 0$. Public targeting would then trigger positive image regard and increase the net utility of participation relative to a setting that conceals this information (where $\phi_i = 0$). This logic corresponds to the standard rationale adopted by many program providers and yields the following null hypothesis:

Hypothesis 0 (*Image regard*). *Informing individuals that they are chosen for a training program because of their group identities leads to positive image views and higher participation than when this information is not disclosed.*

Under H0, take-up and completion rates should be higher in PUBLIC INFO than in NO INFO.

However, public targeting may also threaten the identity of invited students by triggering personal or social image concerns, in which case $\phi_i < 0$. To avoid these costs, invitees may decline the opportunity, lowering participation relative to a no-information setting. This leads to the following alternative hypothesis:

Hypothesis 1 (*Image concern*). *Informing individuals that they are chosen for a training program because of their group identities triggers image concerns that reduce participation compared to when this information is not disclosed.*

Under H1, take-up and completion rates should be higher in NO INFO than in PUBLIC INFO.

I do not have explicit hypotheses for the PRIVATE INFO condition. This treatment serves as a complement to measure which image dimensions may be affected most by the targeting.¹³ I test Hypothesis 0 and Hypothesis 1 in the main field experiment and in a complementary lab-in-the-field experiment reported in Section 4.

The next two hypotheses relax the assumption that program benefits and costs are the same for everyone. On the benefit side, informing individuals that they were selected for the program because of their group identities may change their beliefs about the program's value. Targets may infer that a program offered specifically to certain social groups is of lower quality or relevance, which could reduce take-up even in the absence of image concerns (see e.g., Cronin et al. 2025; Roth et al. 2024). In this case, the benefit term B in Equation (1) becomes a function of ϕ ; identity-based disclosure may lower its perceived value and thus discourage participation (see Hypothesis 2).

Hypothesis 2 (*Program benefits*). *Informing individuals that they are chosen for a training program because of their group identities may reduce the benefits they expect from the program, lowering participation relative to a setting where this information is not disclosed.*

On the cost side, information disclosure may affect the effort required to take up the program, C . In my design, participants must obtain an endorsement from a faculty

¹³ If personal image concerns dominate, PRIVATE INFO and NO INFO outcomes should differ. If social image concerns dominate, PRIVATE INFO and PUBLIC INFO outcomes should differ (Bursztyn and Jensen 2017).

member at the local university. If targeted students fear that endorsers will react negatively once they explicitly reveal that their selection is identity-based, program take-up could fall. In this case, the cost term C in Equation (1) becomes a function of ϕ and rises when identity information is disclosed, increasing the effective cost of registration (see Hypothesis 3).

Hypothesis 3 (*Endorser retaliation*). *Requesting an endorsement that explicitly states that individuals are chosen for a training program because of their group identities may change beliefs about the likelihood of receiving the endorsement or about the endorser’s perceptions, reducing participation relative to a setting where this information is not disclosed in the request.*

To test Hypothesis 2 and Hypothesis 3, I conduct two complementary lab-in-the-field experiments also reported in Section 4.

3 Results

This section presents the main results of the field experiment and establishes the causal impact of identity-based targeting on program participation. The analysis focuses first on take-up to quantify the behavioral barrier introduced by the communication strategy. It then examines completion rates to determine whether the effects persist throughout the duration of the training.¹⁴ Finally, I quantify the welfare costs of the reduced participation by estimating the causal impact of the training program on students’ academic performance. I report two-sided p-values from proportion tests in the main text. These findings are robust to regression specifications that control for individual covariates and targeted social categories, the full details of which are reported in Appendix B.

¹⁴ Take-up was pre-registered as the primary outcome. Completion was not pre-registered, but I include it to assess whether the effects of public targeting operate solely at the registration stage or persist throughout the program (see Appendix H).

3.1 Treatment Effects on Take-up

Pooling data across both waves of the experiment reveals a substantial negative effect of public targeting on participation. Consistent with the hypothesis that identity salience triggers image concerns (H1), explicitly disclosing the selection criteria significantly reduces take-up compared to the condition where this information is concealed. As shown in Figure 1 (Panel A), take-up rates rise from 26% in the PUBLIC INFO condition to 33% in the NO INFO condition ($p < 0.001$).¹⁵ This 27% increase suggests that the strategy of highlighting identity-based eligibility acts as a deterrent rather than as an incentive. The intermediate results for the PRIVATE INFO condition indicate that this dampening effect on program participation is driven by a combination of personal knowledge and the social signal sent to third parties.

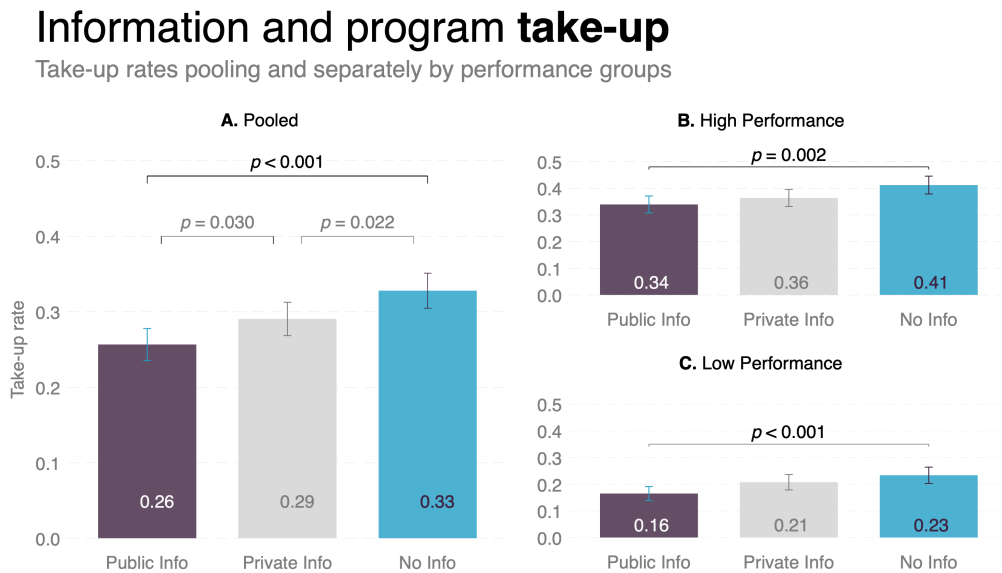


Figure 1 Treatment Effects on Program Take-Up.

Panel A pools data across waves ($N = 4,831$). Panels B and C report results separately for the high-performance (Wave 1) and low-performance (Wave 2) samples. Bars represent participation rates; error bars denote standard errors. P -values are derived from two-sided proportion tests.

The deterrent effect of public targeting proves robust across different academic contexts. As shown in Figure 1 (Panels B and C), the negative impact of the PUBLIC INFO condition is evident in both waves of the experiment. Among high-performing students, concealing the selection criteria increases take-up by 7 percentage points (a 21% in-

¹⁵ For context, Ko and Moffitt (2022) report take-up rates for a wide range of programs, in light of which the average of 29% in my study is comparable to typical rates observed for social benefits.

crease) relative to the public disclosure condition. Among low-performing students it generates a matching 7 percentage point gain (a 44% increase). This indicates that the behavioral aversion to identity-salient targeting operates independently of whether the invitation might be interpreted as a reward for merit or as remedial support. A detailed heterogeneity analysis reported in Appendix D shows that the patterns are also not driven by a particular type of student: the reduction in participation under PUBLIC INFO arises both for students who hold a single targeted identity and for those with multiple identities, and it is observed for key groups such as women, low socio-economic status students, and those from rural origins. This is summarized in Result 1 below:

Result 1 *Publicly informing individuals that they are chosen for a training program because of their group identities reduces take-up relative to a setting in which this information is not disclosed.*

3.2 Treatment Effects on Completion

Completion rates follow a similar pattern to take-up (see Figure 2 Panel A), increasing from 20% to 24% ($p = 0.003$) when information is not disclosed. To pinpoint where the targeting strategy fails, I analyze progression through the training program using a metric of completed steps ranging from 1 (registration) to 10 (completion of the final session). As illustrated in Figure 2 (Panel B), the informational treatments generate a sharp divergence in overall program completion unconditional on take-up (see solid lines).¹⁶ Participants in the NO INFO condition complete an average of 2.75 steps, significantly more than the 2.23 steps completed by those in the PUBLIC INFO condition ($p < 0.001$). However, this gap is driven entirely by the initial decision to register. Conditional on take-up (see dashed lines), there are no significant differences in persistence. Students complete an average of 8.39 steps in NO INFO versus 8.67 steps in PUBLIC INFO ($p = 0.156$).¹⁷

This indicates that once individuals cross the behavioral threshold of acceptance, they are equally likely to finish the training regardless of how they were targeted. The friction

¹⁶ PRIVATE INFO is omitted from the figure for clarity; results lie between the other two conditions as reported in Appendix D.

¹⁷ Results are robust to Holm-Bonferroni corrections for multiple hypothesis testing, as reported in Appendix H.2.

Information and program completion

Completion rates and progression steps

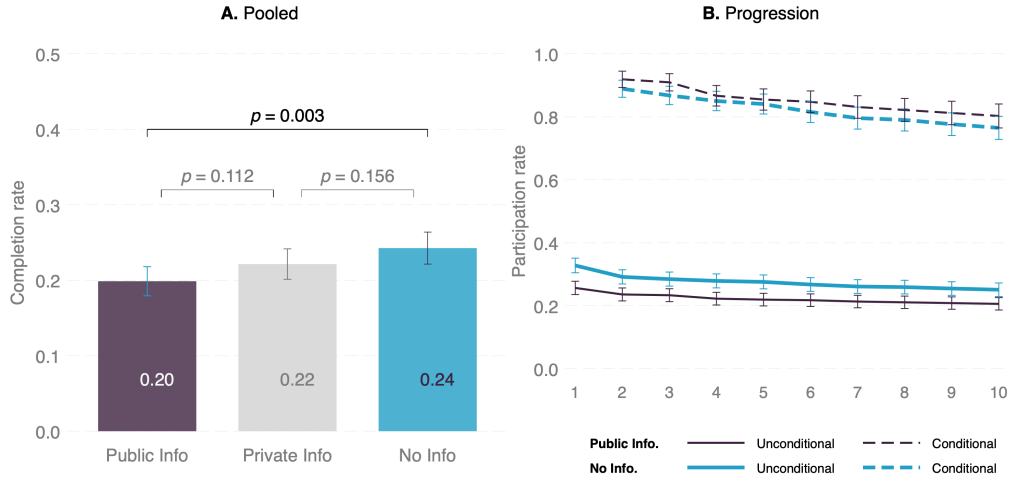


Figure 2 Treatment Effects on Program Completion and Progression.

Panel A reports unconditional completion rates. Panel B plots the average number of steps completed (range 1–10) for the pooled sample (solid lines, unconditional) and the sample restricted to registered students (dashed lines, conditional on take-up). The Private Info condition is omitted from Panel B for visual clarity. Error bars denote standard errors. P -values correspond to two-sided proportion tests (Panel A) and t -tests (Panel B)

imposed by public disclosure is therefore concentrated at the extensive margin. By deterring initial take-up, the targeting strategy prevents a significant share of students from accessing a program they would otherwise complete. This is summarized in Result 2 below:

Result 2 Publicly informing individuals that they are chosen for a training program because of their group identities reduces the average number of completed steps when take-up is considered, but there are no differences across information conditions in the number of steps completed conditional on take-up.

3.3 Welfare Analysis

To quantify the opportunity cost of the reduced participation caused by public targeting, I link the experimental data with administrative academic records covering the five semesters surrounding the intervention for all 4,831 invited students. I estimate the causal impact of the training program on standardized Grade Point Average (GPA) using a difference-in-difference design with student and time fixed effects. Table 2 (Model

I) reports the results and shows that completing the program increases students' standardized GPA by 0.034 standard deviations ($p < 0.001$). Because participation in the training is voluntary, a natural concern is selection on unobservables. To address this, I complement the first analysis with an event study (Model II). The dynamic specification shows that effects are essentially zero in the pre-treatment periods, supporting the parallel trends assumption, and then rise to 0.036, 0.050, and 0.058 standard deviations over the three semesters after the program was rolled out. This shows that students who eventually completed the program were not on a differential positive trajectory prior to the rollout of the training program.

Table 2 Impact of Program Completion on Academic Performance

The table reports estimates from student and semester fixed-effects regressions. The dependent variable is standardized GPA (Z-score). Model I reports static Difference-in-Differences (DiD) estimates. Model II reports event-study coefficients relative to the semester two periods prior to treatment ($t-2$, omitted category). Robust standard errors clustered at the student level are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	I	II
Post $\times$ Completed	0.034*** (0.009)	
Semester ₋₁		0.008 (0.015)
Semester ₀ (<i>rollout</i>)		0.022 (0.015)
Semester ₁		0.036** (0.016)
Semester ₂		0.050*** (0.017)
Semester ₃		0.058*** (0.018)
# Obs.	21641	21641
# Students	4831	4831
FE _{Student}	✓	✓
FE _{Time}	✓	✓

These results make the communication strategy especially consequential. As shown above, the PUBLIC INFO condition lowers take-up by 7 percentage points (a 27% decrease) and completion by 4 percentage points (a 20% decrease) relative to NO INFO. Given the GPA gains for compliers (i.e., those who completed the training program), this drop in participation translates into a loss of human-capital accumulation for the intended beneficiaries. The welfare cost is particularly pronounced because the program

is short, fully online, and delivered at (approximately) zero marginal cost per additional student. In Appendix E.4, I benchmark the effect sizes in relation to different types of interventions (e.g., mentoring or nudging). There I also report heterogeneity analyses showing there are especially strong GPA effects for female and low-SES students, precisely the groups the initiative was designed to support.

4 Mechanisms

While the field experiment establishes the causal impact of the targeting strategy on behavior, it cannot fully disentangle the specific mechanisms driving this behavior. To isolate the psychological mechanisms driving the behavioral response and pinpoint the specific friction introduced by identity-based disclosure, I conducted three complementary lab-in-the-field experiments in the fall of 2024.¹⁸ These studies involved 1,113 students from the same university who were not part of the original intervention, to prevent contamination or priming effects. Across studies, participants were randomly assigned to either a NO INFO or an INFO condition that mirror the field experiment, allowing me to elicit their beliefs and test whether the reduction in participation stems from negative inferences about program quality, fears of endorser retaliation, or image concerns.¹⁹

Experiment 1: Perceived Program Benefits. I first test whether the targeting strategy signals negative information about the value of the opportunity. If students infer that a program restricted to specific demographic groups is of lower quality or relevance, this lower perceived benefit would reduce participation, as proposed in Hypothesis 2 (see e.g., Cronin et al. 2025; Roth et al. 2024). Data from an experiment with 401 students provide no support for this channel. As shown in Figure 3.A, participants in both the INFO and NO INFO conditions report high expected benefits from the training, with no statistical difference between the two groups (ratings of 8.88 vs. 8.89 out of 10, $p = 0.897$). Explicitly linking the offer to group identities does not appear to dilute the

¹⁸ These experiments were pre-registered in the AEA RCT Registry 0014285, 0014810, and 0014736. See Appendix G for the timeline.

¹⁹ While a revealed-preference design with real program offers would ideally test these mechanisms, it was ethically infeasible. Given that the main field experiment had already established that public targeting significantly reduces participation, it was unethical to expose a new cohort to this expected barrier. Consequently, I rely on belief elicitations to investigate the mechanisms.

Information and mechanisms

Perceived benefits, endorser reactions, and image concerns

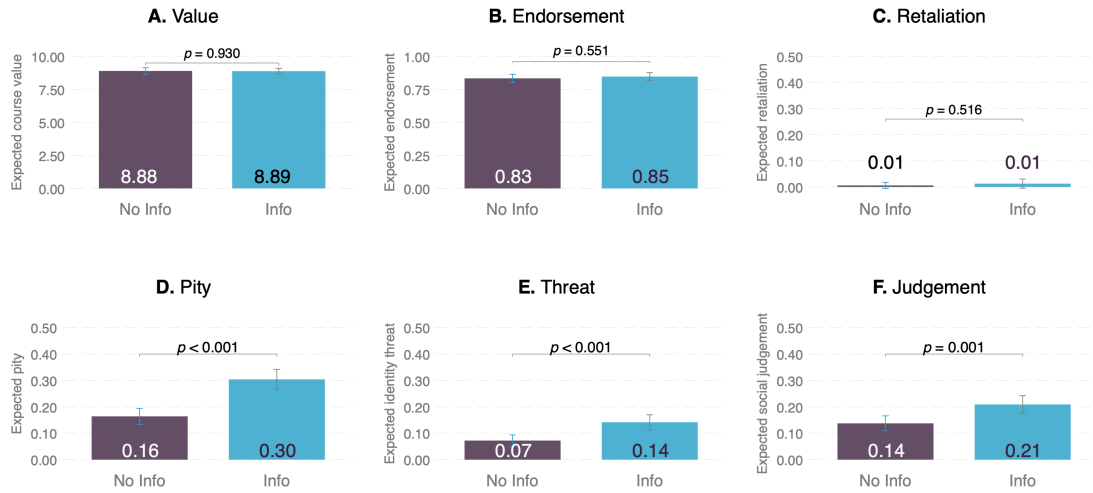


Figure 3 Mechanisms: Belief Elicitation Results.

The figure reports mean belief scores from three complementary online experiments. Panel A: Perceived program value (0–10 scale). Panel B: Expected endorsement probability (0–1). Panel C: Expected endorser retaliation (binary). Panels D–F: Expected feelings of pity, stereotype threat, and social judgment resulting from the invitation. Error bars denote standard errors. *P*-values correspond to two-sided *t*-tests comparing the Info and No Info conditions.

perceived value of the program. Moreover, the beliefs are aligned with the very positive course evaluations of the training program completed by 1,063 of the 1,066 students who finished all sessions (see Appendix F). This indicates that the belief elicitation are consistent with comparable field experimental metrics.

Experiment 2: Strategic Considerations regarding Endorsers. I similarly find no evidence that the results are driven by strategic concerns regarding the faculty endorsement (Hypothesis 3). In a second experiment with 330 students, participants reviewed the specific endorsement script required for their condition. As reported in Figure 3 (Panel B), beliefs about the endorser’s reaction were overwhelmingly positive in both conditions. Over 80% of students expected their request for an endorsement to be granted ($p = 0.551$). These beliefs align with the administrative data from the field, where 97% of endorsement requests were granted. Furthermore, only three out of 330 students (0.9%) believe the endorser would perceive them or treat them more negatively afterward (Figure 3 Panel C). The barrier to participation is therefore not a rational calculation regarding program quality or transaction risks.

Experiment 3: Image Concerns. In stark contrast to the null results on program value and endorser retaliation, the data provide strong support for the image concern hypothesis (Hypothesis 1). In a third experiment with 382 students, I measured second-order beliefs regarding the psychological cost of the invitation.²⁰ The results indicate that the targeting strategy fundamentally alters the social meaning of the offer (see Figure 3 Panels C, D and E). Respondents believe that receiving an invitation explicitly tied to their group identity triggers significantly higher feelings of pity (30% vs. 16%, $p < 0.001$), stereotype threat (14% vs. 7%, $p < 0.001$), and social judgment (14% vs. 21%, $p = 0.001$) compared to the condition where eligibility is not disclosed (i.e., NO INFO).²¹

The disclosure of identity-based selection transforms the invitation into a negative signal, indicating that the behavioral withdrawal observed in the field is driven by a desire to avoid these image costs. Collectively, these findings rule out instrumental explanations and pinpoint the mechanism: the invitation imposes a psychological tax that outweighs the material benefits of the program. This is summarized in the following result:

Result 3 Evidence from three complementary experiments shows that the negative effect of public targeting is not driven by perceived low program benefits or anticipated endorser reactions; instead, explicitly disclosing that selection is based on group identities triggers image concerns that reduce participation.

5 Conclusion

This paper demonstrates that the communication strategy used to implement diversity initiatives can fundamentally undermine their effectiveness. While institutions often highlight identity-based eligibility to signal a commitment to inclusion, my results show that this practice can generate a powerful deterrent effect. In a large-scale natural field experiment, explicitly disclosing that selection was based on demographic criteria reduced program take-up by 27% and completion by 20% relative to a condition where

²⁰ Several channels may give rise to image concerns, including fear of judgement (List 2004), stereotype threat (Steele and Aronson 1995), or aversion to pity (Leibbrandt and List 2018), so I test each of them in the experiment.

²¹ The text reports the combined measure; disaggregated results by group identity are reported in Appendix G.4.

eligibility criteria were concealed. This behavioral response generates a significant efficiency loss. Because the training program causally increases academic performance, the targeting strategy effectively discourages the intended beneficiaries from accessing a high-return opportunity. The method of communication thus undermines the very objective of the policy.

The experimental evidence pinpoints the mechanism driving this friction. It is not that targeted students perceive the program as less valuable or fear rejection from endorsers. Rather, the barrier is psychological. As shown by the complementary belief elicitation, making identity salient transforms the invitation into a signal of anticipated pity, stereotype threat, and social judgment. Beneficiaries are therefore willing to forgo the material benefits of the program to avoid the image cost associated with public disclosure. This result highlights that the psychological costs of take-up are not fixed parameters but are endogenous to how the opportunity is framed. By attaching an identity label to the offer, the institution inadvertently imposes a psychological tax that filters out the students it aims to help.

These results contribute to the literature on the design of affirmative action and diversity policies. While extensive research focuses on the definition of eligibility rules and admissions criteria, this study highlights the critical role of implementation and communication. I show that even when a program is free, universally accessible, and incentivized, the specific way in which these policies are communicated to potential beneficiaries can operate as a gatekeeper. By identifying the behavioral costs of making identity salient, this paper helps explain why eligible individuals frequently leave benefits on the table. It suggests that the incomplete take-up observed in many social programs may stem not only from structural barriers or information deficits but from the image costs implicit in the targeting design itself.

From a policy perspective, the findings suggest a concrete reform to the design of inclusion initiatives. Institutions should continue to use administrative data to identify and prioritize specific groups for support, but they should decouple this administrative definition from the public communication of the offer. By stripping the invitation of identity-based labels, as mirrored by the no-information condition in this study, policymakers can ensure resources reach the intended demographic groups without triggering the image concerns that suppress participation. This approach shifts the burden

of identification from the individual to the institution and thereby reconciles the goal of equity with efficiency. Ultimately, successful inclusion requires designing communication strategies that respect the social image of beneficiaries, ensuring that the act of accepting valuable opportunities does not come at the cost of dignity.

References

- Alan, S., Boneva, T., and Ertac, S. (2019). Ever failed, try again, succeed better: Results from a randomized education intervention on grit. Quarterly Journal of Economics, 134:1121–1162.
- Alan, S. and Ertac, S. (2018). Fostering patience in the classroom: Results from randomized educational intervention. Journal of Political Economy, 126:1865–1911.
- Athey, S. and Palikot, E. (2024). The value of non-traditional credentials in the labor market. Working Paper 4189, Stanford Graduate School of Business.
- Bearson, D. F. and Sunstein, C. R. (2025). Take up. Behavioral Public Policy, 9:849–864.
- Bertrand, M. and Duflo, E. (2017). Field experiments on discrimination. In Banerjee, A. V. and Duflo, E., editors, Handbook of Economic Field Experiments, volume 1, chapter 8, pages 309–393. Elsevier.
- Bertrand, M., Hanna, R., and Mullainathan, S. (2010). Affirmative action in education: Evidence from engineering college admissions in india. Journal of Public Economics, 94:16–29.
- Beshears, J., Lee, H. N., Milkman, K. L., Mislavsky, R., and Wisdom, J. (2021). Creating exercise habits using incentives: The trade-off between flexibility and routinization. Management Science, 67(7):3985–4642.
- Beshears, J., Milkman, K. L., and Schwartzstein, J. (2016). Beyond beta-delta: The emerging economics of personal plans. American Economic Review (Papers & Proceedings), 106(5):430–434.
- Bettinger, E. P. and Baker, R. B. (2014). The effects of student coaching: An evaluation

- of a randomized experiment in student advising. Educational Evaluation and Policy Analysis, 36(1):3–19.
- Bhargava, S. and Manoli, D. (2015). Psychological frictions and the incomplete take-up of social benefits: Evidence from an irs field experiment. American Economic Review, 105(11):3489–3529.
- Bogliacino, F., Jiménez Lozano, L., and Reyes, D. (2018). Socioeconomic stratification and stereotyping: Lab-in-the-field evidence from colombia. International Review of Economics, 65(1):77–118.
- Burland, E., Dynarski, S., Micheltmore, K., Owen, S., and Raghuraman, S. (2023). The power of certainty: Experimental evidence on the effective design of free tuition programs. American Economic Review: Insights, 5(3):293–310.
- Bursztyn, L., Callen, M., Ferman, B., Gulzar, S., Hasanain, A., and Yuchtman, N. (2020). Political identity: Experimental evidence on anti-americanism in pakistan. Journal of the European Economic Association, 18(5):2532–2560.
- Bursztyn, L., Egorov, G., and Jensen, R. (2019). Cool to be smart or smart to be cool? understanding peer pressure in education. Review of Economic Studies, 86(4):1487–1526.
- Bursztyn, L., Fujiwara, T., and Pallais, A. (2017). 'acting wife': Marriage market incentives and labor market investments. American Economic Review, 107(11):3288–3319.
- Bursztyn, L. and Jensen, R. (2015). How does peer pressure affect educational investments? Quarterly Journal of Economics, 130(3):1329–1367.
- Bursztyn, L. and Jensen, R. (2017). Social image and economic behavior in the field: Identifying, understanding, and shaping social pressure. Annual Review of Economics, 9:131–153.
- Butera, L., Metcalfe, R., Morrison, W., and Taubinsky, D. (2022). Measuring the welfare effects of shame and pride. American Economic Review, 112(1):122–168.

- Cárdenas, J. C., Fergusson, L., and García Villegas, M. (2021). La quinta puerta: De cómo la educación en Colombia agudiza las desigualdades en lugar de remediarlas. Ariel (Editorial Planeta Colombiana), Bogotá, Colombia.
- Carlana, M. and Fort, M. (2022). Hacking gender stereotypes: Girls' participation in coding clubs. AEA Papers and Proceedings, 112:583–587.
- Carlana, M. and La Ferrara, E. (2024). Apart but connected: Online tutoring, cognitive outcomes, and soft skills. Working Paper 32272, National Bureau of Economic Research.
- Carlana, M., La Ferrara, E., and Pinotti, P. (2022). Goals and gaps: Educational careers of immigrant children. Econometrica, 90(1):1–29.
- Cronin, C. J., Forsström, M. P., and Papageorge, N. W. (2025). What good are treatment effects without treatment? mental health and the reluctance to use talk therapy. Review of Economic Studies, 92(3):1699–1737.
- DellaVigna, S., List, J. A., and Malmendier, U. (2012). Testing for altruism and social pressure in charitable giving. Quarterly Journal of Economics, 127(1):1–56.
- DellaVigna, S., List, J. A., Malmendier, U., and Rao, G. (2017). Voting to tell others. Review of Economic Studies, 84(1):143–181.
- Dynarski, S., Libassi, C. J., Micheltore, K., and Owen, S. (2021). Closing the gap: The effect of reducing complexity and uncertainty in college pricing on the choices of low-income students. American Economic Review, 111(8):1721–1756.
- Finkelstein, A. and Notowidigdo, M. J. (2019). Take-up and targeting: Experimental evidence from snap. Quarterly Journal of Economics, 134(3):1505–1556.
- Fryer, R. G., Loury, G. C., and Yuret, T. (2008). An economic analysis of color-blind affirmative action. Journal of Law, Economics & Organization, 24(2):319–355.
- Harrison, G. W. and List, J. A. (2004). Field experiments. Journal of Economic Literature, 42(4):1009–1055.
- Ko, W. and Moffitt, R. A. (2022). Take-up of social benefits. Working Paper 30148, National Bureau of Economic Research.

- Leibbrandt, A. and List, J. A. (2018). Do equal employment opportunity statements backfire? evidence from a natural field experiment on job-entry decisions. Working Paper 25035, National Bureau of Economic Research.
- List, J. A. (2004). The nature and extent of discrimination in the marketplace: Evidence from the field. Quarterly Journal of Economics, 119(1):49–89.
- Londoño Vélez, J. (2022). The impact of diversity on perceptions of income distribution and preferences for redistribution. Journal of Public Economics, 214:104732.
- Milkman, K. L. (2021). How to Change: The Science of Getting from Where You Are to Where You Want to Be. Penguin Random House, New York.
- Moffitt, R. (1983). An economic model of welfare stigma. American Economic Review, 73(5):1023–1035.
- Oreopoulos, P. and Petronijevic, U. (2018). Student coaching: How far can technology go? The Journal of Human Resources, 53(2):299–329.
- Roth, C., Schwardmann, P., and Tripodi, E. (2024). Misperceived effectiveness and the demand for psychotherapy. Journal of Public Economics, 240:105254.
- Sacerdote, B. (2001). Peer effects with random assignment: Results from dartmouth roommates. Quarterly Journal of Economics, 116(2):681–704.
- Shampanier, K., Mazar, N., and Ariely, D. (2007). Zero as a special price: The true value of free products. Marketing Science, 26(6):742–757.
- Steele, C. M. and Aronson, J. (1995). Stereotype threat and the intellectual test performance of african-americans. Journal of Personality and Social Psychology, 69(5):797–811.
- Tversky, A. and Kahneman, D. (1986). Rational choice and the framing of decisions. Journal of Business, 59(4):S251–S278.

Online Appendix:

Publicly targeting group identities impacts take-up of educational opportunities

Manuel Munoz

A Invitation Email

This appendix presents the experimental materials and validates the integrity of the randomization. First, I reproduce the invitation message sent to all eligible students in Spanish by the Office of International Relations of the local university. I provide the English translation (by the author) and indicate experimental variations in content with the labels NO INFO, PRIVATE INFO, and PUBLIC INFO. Second, I report the balance of covariates across treatments (Table A-1), demonstrating that the randomization resulted in comparable samples across conditions.

A.1 Message to students

Dear [Student Name],

The [Local University] has a collaboration agreement with [International University], a globally recognized university. As part of this agreement, professors from [International University] will teach a training program for students at [Local University] to acquire and further develop tools to achieve their goals and increase their chances of personal and professional success (details about the program appear at the end of this message).

This great opportunity provides several benefits: the chance to learn from excellent professors, an attendance certificate from [International University]. These types of certificates can significantly strengthen your CV and open doors to future jobs or scholarships. In addition, at the end of the program there will be a lottery for several

recent-generation iPads among those who complete the program, to provide students with a tool that may help in their academic endeavors.

- NO INFO: You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA].
- PRIVATE INFO or PUBLIC INFO: You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA] and because you fulfill one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a first-generation student (neither of your parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

In order to register for the program and indicate you are interested in benefiting from this great opportunity, you will need to follow two very simple steps:

1. Pre-registration: Respond to this message indicating your interest in taking part in the program. This will count as a pre-registration.
2. Endorsement from a faculty member at [Local University]: Send an email message to a faculty member, copying [Program's Email Address] (cc), and ask them to reply with the following message:
 - NO INFO or PRIVATE INFO: I, [Professor's name] endorse student [Your Name] to take part in the training program "How to change: scientific tools to achieve the goals in your personal and professional life", because he/she can benefit from this program, as his/her cumulative GPA is [Student's GPA].
 - PUBLIC INFO: I, [Professor's name] endorse student [Your Name] to take part in the training program "How to change: scientific tools to achieve the goals in your personal and professional life", because he/she can benefit from this program, as his/her cumulative GPA is [Student's GPA], and because he/she fulfills at least one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a

first-generation student (neither of his/her parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

Once the faculty member has replied, you will be officially registered. It is essential that both you and the professor include this email address in copy for all communications.

All professors at [Local University] have been informed about this great opportunity and are willing to help you with the required endorsement.

Additional information about the program:

- Name: “How to change: scientific tools to achieve the goals in your personal and professional life”.
- Instructor(s): The program will be taught by professors of high international standing from the [International University].
- Language: Spanish.
- Duration: 9 online sessions, half an hour each. All sessions are independent and you will be able to complete them at your own pace. So, you will not have any conflicts of scheduling with other academic activities.
- Start: The program will start on [Start date].
- Costs: Free course.
- Requirements: To be pre-selected and to be endorsed by a faculty member from the [Local University].
- Benefits: An international certificate of attendance. Also, you will participate in the lottery of various iPads.
- Registration deadline: Please pre-register before [Deadline date].

We await your positive response so you can benefit from the opportunities in this program.

Sincerely,

[Signature]

Head / Office of International Relations

A.2 Randomization Balance

Within each wave of the study, I randomly assigned students to one of the three information conditions (PUBLIC INFO, PRIVATE INFO, NO INFO) using block randomization based on the targeted categories. This procedure yields balanced samples across treatments in both waves, as illustrated in Table A-1 below:

Table A-1 Balance of Covariates Across Treatments

Columns I–III and V–VII report the mean of each covariate with standard errors in parentheses. Columns IV and VIII report p -values from an ANOVA test of equality of means across the three treatments for the high- and low-performance waves, respectively.

	High Performance				Low Performance			
	PUBLIC INFO I	PRIVATE INFO II	NO INFO III	p-value IV	PUBLIC INFO V	PRIVATE INFO VI	NO INFO VII	p-value VIII
Female	0.69 (0.46)	0.68 (0.46)	0.66 (0.47)	0.43	0.62 (0.48)	0.62 (0.48)	0.58 (0.49)	0.18
Rural origins	0.34 (0.47)	0.38 (0.48)	0.38 (0.48)	0.22	0.29 (0.45)	0.29 (0.45)	0.33 (0.47)	0.12
Low SES	0.92 (0.26)	0.92 (0.26)	0.92 (0.27)	0.81	0.87 (0.32)	0.90 (0.29)	0.90 (0.29)	0.21
First generation	0.14 (0.35)	0.15 (0.36)	0.16 (0.37)	0.52	0.12 (0.33)	0.12 (0.33)	0.14 (0.35)	0.47
Ethnic minority	0.02 (0.15)	0.02 (0.15)	0.02 (0.14)	0.78	0.03 (0.17)	0.02 (0.16)	0.03 (0.18)	0.79
Observations	864	864	833		776	757	737	

B Regression Results

This appendix provides the formal econometric estimation of the treatment effects reported in Section 3. To quantify the impact of information disclosure on program participation while controlling for individual covariates, I estimate the following Linear Probability Model:

$$Y_i = \alpha + \beta_1 \text{PUBLIC INFO}_i + \beta_2 \text{PRIVATE INFO}_i + \gamma \mathbf{X}_i + \varepsilon_i \quad (\text{B-1})$$

where Y_i is the binary outcome variable (taking the value of 1 for Take-up or Completion) for student i . The omitted category is the NO INFO condition, meaning the coefficients β_1 and β_2 capture the change in participation relative to the condition of no information disclosure. The vector $\mathbf{X}_i$ includes the full set of stratification controls used in the block randomization: gender, socio-economic status, rural origin, ethnicity, and first-generation status. Standard errors are robust to heteroskedasticity.

Table B-1 pools both performance groups. Tables B-2 and B-3 restrict the sample to high- and low-performance students (first and second waves), respectively. Tables B-4 and B-5 focus on targeted students with multiple versus single group identities. Tables B-6, B-7, B-8, B-9, and B-10 report results for subgroups defined by each targeted identity in turn: female, rural origin, low socio-economic status, first-generation, and ethnic minority students. In all of these subgroup tables, the condition is that a targeted student holds the relevant identity, irrespective of any additional identities. Because some administrative profiles lack information for at least one demographic control, I impute missing values using the sample mean of each variable.

For all tables, the dependent variable is the rate of take-up in columns I-II, the completion rate in columns III-IV, and the number of completed steps (where take-up is step 1 and the ninth session is step 10) in columns V-VI (unconditional on take-up) and in columns VII-VIII (conditional on take-up).

Table B-1 Treatment Effects on Participation (Pooled Sample).

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	−0.071*** (0.016)	−0.074*** (0.016)	−0.044*** (0.015)	−0.045*** (0.015)	−0.528*** (0.148)	−0.544*** (0.147)	0.275 (0.194)	0.275 (0.195)
PRIVATE INFO	−0.037** (0.016)	−0.040** (0.016)	−0.021 (0.015)	−0.023 (0.015)	−0.307** (0.151)	−0.324** (0.150)	0.024 (0.199)	0.046 (0.198)
Constant	0.328*** (0.012)	0.230*** (0.029)	0.243*** (0.011)	0.170*** (0.026)	2.753*** (0.109)	1.938*** (0.263)	8.392*** (0.136)	8.499*** (0.352)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	4831	4831	4831	4831	4831	4831	1407	1407
R^2	0.004	0.025	0.002	0.014	0.003	0.020	0.002	0.004
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.030	0.028	0.112	0.110	0.126	0.124	0.210	0.252

Table B-2 Treatment Effects on Participation (High Performance).

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	-0.073*** (0.023)	-0.076*** (0.023)	-0.034 (0.022)	-0.036* (0.022)	-0.479** (0.220)	-0.503** (0.219)	0.392* (0.234)	0.382 (0.236)
PRIVATE INFO	-0.048** (0.024)	-0.048** (0.023)	-0.020 (0.022)	-0.020 (0.022)	-0.360 (0.221)	-0.357 (0.219)	0.128 (0.242)	0.152 (0.240)
Constant	0.412*** (0.017)	0.365*** (0.043)	0.305*** (0.016)	0.268*** (0.041)	3.466*** (0.159)	3.057*** (0.408)	8.417*** (0.168)	8.485*** (0.420)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	2561	2561	2561	2561	2561	2561	950	950
R ²	0.004	0.035	0.001	0.018	0.002	0.026	0.003	0.010
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.290	0.215	0.519	0.441	0.585	0.496	0.268	0.332

Table B-3 Treatment Effects on Participation (Low Performance).

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	-0.068*** (0.021)	-0.068*** (0.020)	-0.054*** (0.018)	-0.054*** (0.018)	-0.571*** (0.185)	-0.569*** (0.184)	0.001 (0.349)	0.015 (0.354)
PRIVATE INFO	-0.026 (0.021)	-0.026 (0.021)	-0.023 (0.019)	-0.023 (0.019)	-0.255 (0.193)	-0.259 (0.192)	-0.184 (0.348)	-0.187 (0.350)
Constant	0.233*** (0.016)	0.131*** (0.035)	0.172*** (0.014)	0.103*** (0.031)	1.947*** (0.141)	1.146*** (0.312)	8.343*** (0.233)	8.588*** (0.638)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	2270	2270	2270	2270	2270	2270	457	457
R ²	0.005	0.022	0.004	0.014	0.004	0.018	0.001	0.004
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.033	0.034	0.078	0.083	0.076	0.080	0.616	0.587

Table B-4 Treatment Effects for Multiple-Identity Holders.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	-0.075*** (0.020)	-0.078*** (0.020)	-0.039** (0.019)	-0.042** (0.019)	-0.509*** (0.189)	-0.538*** (0.189)	0.427* (0.219)	0.424* (0.221)
PRIVATE INFO	-0.042** (0.021)	-0.045** (0.020)	-0.032* (0.019)	-0.034* (0.019)	-0.392** (0.189)	-0.418** (0.188)	-0.126 (0.233)	-0.118 (0.233)
Constant	0.358*** (0.015)	0.135* (0.076)	0.266*** (0.014)	0.089 (0.066)	3.016*** (0.138)	1.076 (0.681)	8.436*** (0.157)	8.224*** (0.958)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	3179	3179	3179	3179	3179	3179	1012	1012
R ²	0.004	0.026	0.002	0.015	0.003	0.021	0.006	0.007
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.096	0.087	0.695	0.674	0.524	0.507	0.016	0.018

Table B-5 Treatment Effects for Single-Identity Holders.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	-0.063*** (0.026)	-0.067*** (0.026)	-0.051** (0.023)	-0.053** (0.023)	-0.544** (0.231)	-0.570** (0.231)	-0.106 (0.399)	-0.015 (0.389)
PRIVATE INFO	-0.032 (0.027)	-0.037 (0.026)	-0.002 (0.019)	-0.005 (0.019)	-0.162 (0.245)	-0.202 (0.244)	0.417 (0.377)	0.537 (0.379)
Constant	0.271*** (0.019)	0.365*** (0.043)	0.199*** (0.017)	0.304*** (0.039)	2.247*** (0.175)	3.408*** (0.390)	8.281*** (0.271)	10.051*** (0.653)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	1652	1652	1652	1652	1652	1652	395	395
R ²	0.004	0.015	0.004	0.012	0.004	0.015	0.005	0.047
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.209	0.216	0.029	0.035	0.095	0.108	0.185	0.156

Table B-6 Treatment Effects: Female Students.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	−0.089*** (0.021)	−0.060** (0.028)	−0.051*** (0.019)	−0.049** (0.025)	−0.627*** (0.190)	−0.525** (0.250)	0.050 (0.034)	−0.024 (0.063)
PRIVATE INFO	−0.033 (0.021)	−0.007 (0.029)	−0.017 (0.019)	−0.015 (0.026)	−0.264 (0.194)	−0.154 (0.259)	0.021 (0.033)	−0.041 (0.060)
Constant	0.368*** (0.015)	0.255*** (0.021)	0.266*** (0.014)	0.188*** (0.019)	3.046*** (0.141)	2.142*** (0.191)	0.722*** (0.024)	0.736*** (0.042)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	3134	3134	3134	3134	3134	3134	1022	1022
R ²	0.006	0.038	0.002	0.027	0.004	0.033	0.002	0.007
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.004	0.046	0.064	0.151	0.048	0.120	0.383	0.777

Table B-7 Treatment Effects: Rural-Origin Students.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	−0.062** (0.027)	−0.059 (0.038)	−0.045* (0.025)	−0.038 (0.034)	−0.480* (0.248)	−0.384 (0.345)	0.011 (0.049)	0.031 (0.085)
PRIVATE INFO	−0.041 (0.027)	−0.042 (0.039)	−0.050** (0.024)	−0.064* (0.033)	−0.438* (0.244)	−0.506 (0.338)	−0.069 (0.050)	−0.155* (0.092)
Constant	0.299*** (0.019)	0.251*** (0.028)	0.232*** (0.018)	0.186*** (0.025)	2.559*** (0.178)	2.061*** (0.249)	0.775*** (0.032)	0.742*** (0.056)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	1650	1650	1650	1650	1650	1650	438	438
R ²	0.003	0.011	0.003	0.014	0.003	0.014	0.007	0.019
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.427	0.654	0.860	0.426	0.861	0.711	0.134	0.055

Table B-8 Treatment Effects: Low-SES Students.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	-0.069*** (0.018)	-0.084*** (0.024)	-0.039** (0.017)	-0.063*** (0.022)	-0.499*** (0.171)	-0.702 (0.220)	0.047 (0.030)	0.004 (0.058)
PRIVATE INFO	-0.043** (0.019)	-0.052** (0.025)	-0.029* (0.017)	-0.044** (0.022)	-0.400** (0.171)	-0.514** (0.224)	0.013 (0.030)	-0.025 (0.056)
Constant	0.340*** (0.013)	0.249*** (0.018)	0.256*** (0.012)	0.188*** (0.016)	2.898*** (0.125)	2.121*** (0.167)	0.751*** (0.021)	0.757*** (0.036)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	3799	3799	3799	3799	3799	3799	1150	1150
R ²	0.004	0.041	0.002	0.031	0.003	0.037	0.002	0.004
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.157	0.163	0.532	0.359	0.548	0.361	0.277	0.639

Table B-9 Treatment Effects: First-Generation Students.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps Unconditional		Completed steps Conditional	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	0.027 (0.045)	-0.026 (0.061)	0.047 (0.042)	-0.042 (0.054)	0.579 (0.420)	-0.184 (0.546)	0.070 (0.065)	-0.092 (0.126)
PRIVATE INFO	-0.004 (0.044)	0.030 (0.063)	0.028 (0.041)	0.013 (0.057)	0.269 (0.412)	0.322 (0.576)	0.082 (0.065)	-0.034 (0.116)
Constant	0.340*** (0.013)	0.249*** (0.018)	0.268*** (0.028)	0.204*** (0.039)	3.061*** (0.282)	2.204*** (0.388)	0.702*** (0.047)	0.759*** (0.080)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	704	704	704	704	704	704	274	274
R ²	0.001	0.051	0.002	0.043	0.003	0.051	0.007	0.019
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.495	0.374	0.652	0.328	0.474	0.378	0.843	0.655

Table B-10 Treatment Effects: Ethnic Minority Students.

OLS estimates. The dependent variable is listed in the column header. Take-up and Completion are binary; Completed steps is continuous (1–10). PUBLIC INFO and PRIVATE INFO Info are indicators relative to the omitted NO INFO category. Even-numbered columns include controls for gender, SES, rural origin, ethnicity, and first-generation status. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Take-up		Completion		Completed steps <i>Unconditional</i>		Completed steps <i>Conditional</i>	
	I	II	III	IV	V	VI	VII	VIII
PUBLIC INFO	−0.088 (0.090)	−0.069 (0.121)	−0.107 (0.085)	−0.111 (0.108)	−0.945 (0.854)	−0.865 (1.082)	−0.139 (0.166)	−0.257 (0.278)
PRIVATE INFO	0.150 (0.104)	0.140 (0.139)	−0.018 (0.095)	0.087 (0.132)	0.300 (0.942)	1.297 (1.327)	−0.361* (0.148)	−0.079 (0.208)
Constant	0.279*** (0.069)	0.269*** (0.089)	0.256*** (0.067)	0.231*** (0.085)	2.605*** (0.670)	2.385*** (0.841)	0.917*** (0.083)	0.857*** (0.144)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
# Obs.	132	132	132	132	132	132	39	39
R ²	0.046	0.047	0.013	0.031	0.017	0.032	0.124	0.289
<i>P-values of differences</i>								
NO INFO vs. PRIVATE INFO	0.015	0.123	0.294	0.105	0.144	0.082	0.247	0.532

C Peer Spillovers

C.1 The measure

This appendix examines whether the individual-level randomization in the field experiment generated spillovers through peer communication. For this, I reconstruct the students' social network using administrative course-enrollment records. This proxy is particularly informative in this setting. Because the local university is non-residential (students do not live on campus) and follows a rigid curriculum with small class sizes (typically 20-30 students), shared classrooms constitute the primary mechanism for systematic peer interaction. Furthermore, the program was not publicly advertised and the only communication consisted of the individualized invitation emails (see Section 2.3), limiting information diffusion to direct social contacts.

I construct the network using registration data from the six academic periods preceding the intervention. Two students are defined as peers if they shared at least one course section during this window, with tie strength measured by the number of shared courses. I successfully link 90.7% of the invited students to the network. The remaining 9.3% are not found in the co-enrollment records relative to the invited sample, likely because they did not share classes with any other targeted students during the observation window (i.e., network isolates). This procedure yields several peer measures, including degree, weighted degree, and peer-treatment composition, which I use to assess whether exposure to treated classmates generated spillovers across experimental conditions.

C.2 Empirical Strategy

To quantify the impact of peer exposure on program participation, I estimate Linear Probability Models consistent with the main analysis reported in Equation B-1 in Appendix B. In this case, Y_i denotes the binary take-up decision for student i . I also extend the baseline specification to include measures of peer exposure and their interaction with the treatment indicators. I estimate models of the form:

$$\begin{aligned}
Y_i = & \alpha + \beta_1 \text{PUBLIC INFO}_i + \beta_2 \text{PRIVATE INFO}_i + \gamma \kappa_i \\
& + \lambda_1 (\text{PUBLIC INFO}_i \times \kappa_i) + \lambda_2 (\text{PRIVATE INFO}_i \times \kappa_i) + \epsilon_i
\end{aligned}
\tag{B-2}$$

where the NO INFO condition serves as the omitted category. The variable κ_i represents the measure of peer exposure for student i . The coefficients of interest are the interaction terms λ_1 and λ_2 , which test whether the treatment effects estimated in the main analysis are conditional on the student's peer environment.

I employ two distinct definitions for κ_i to capture different spillover channels. First, I focus on *network degree* (κ_i). It refers to the total number of invited peers with whom student i shares classes. This tests whether the treatment effect varies with the total social density around the student. Second, I focus on *cross-treatment exposure* (κ_i^{diff}). It refers to the number of co-enrolled peers assigned to a *different* treatment condition than student i . This constitutes a direct test of the Stable Unit Treatment Value Assumption (SUTVA), as significant estimates would indicate that a student's response is altered by exposure to conflicting information.

I estimate these models using Ordinary Least Squares with robust standard errors (see Table D-1). To ensure robustness, I replicate the analysis across three network definitions based on tie strength (sharing ≥ 1 , ≥ 5 , or ≥ 10 courses). Finally, to explore potential spillovers within more homogeneous academic environments, I estimate these specifications separately by academic division (see Table D-2). There are 6 divisions at the local university which group related programs: *Health, Law, Humanities, Economics, Engineering* and *Technical programs*. For this division-level analysis, I run the full set of network models but report results only for the strong-tie network (at least ten shared courses between students), noting that the conclusions are unchanged across all network definitions.

C.3 Results

Table D-1 presents the results of the spillover analysis. The estimation yields three key patterns that collectively rule out peer diffusion as a driver of the treatment effects.

Table D-1 Peer Spillovers in the Co-Enrollment Network.

OLS estimates of program take-up. κ_i denotes the number of co-enrolled peers; κ_i^{diff} denotes the number of peers in a different treatment. Columns correspond to networks defined by sharing ≥ 1 , ≥ 5 , or ≥ 10 courses. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	I	II	III	IV	V	VI	VII	VIII	IX
PUBLIC INFO	-0.072** (0.017)	-0.072** (0.017)	-0.064* (0.029)	-0.073** (0.017)	-0.073* (0.029)	-0.061* (0.024)	-0.069** (0.024)	-0.045+ (0.026)	-0.048+ (0.026)
PRIVATE INFO	-0.039* (0.017)	-0.038* (0.017)	-0.029 (0.029)	-0.039* (0.017)	-0.027 (0.029)	-0.043+ (0.025)	-0.043+ (0.025)	-0.041 (0.027)	-0.040 (0.026)
κ_i		-0.000** (0.000)	-0.000 (0.000)			-0.001** (0.000)		-0.002* (0.001)	
PUBLIC INFO $\times$ κ_i			-0.000 (0.000)			-0.001 (0.001)		-0.002+ (0.001)	
PRIVATE INFO $\times$ κ_i			-0.000 (0.000)			-0.000 (0.001)		0.000 (0.001)	
κ_i^{diff}				-0.001** (0.000)	-0.000 (0.000)		-0.002** (0.001)		-0.003* (0.001)
PUBLIC INFO $\times$ κ_i^{diff}					0.000 (0.000)		-0.000 (0.001)		-0.002+ (0.001)
PRIVATE INFO $\times$ κ_i^{diff}					-0.000 (0.000)		0.000 (0.001)		0.000 (0.001)
Constant	0.323** (0.012)	0.350** (0.016)	0.344** (0.022)	0.350** (0.016)	0.346** (0.021)	0.362** (0.018)	0.363** (0.018)	0.367** (0.019)	0.366** (0.019)
# Obs.	4383	4383	4383	4383	4383	3959	3959	3451	3451
R^2	0.004	0.006	0.006	0.006	0.006	0.012	0.011	0.014	0.013

First, the direct negative effect of public targeting is robust to controlling for the peer environment. Across all specifications (Columns I–IX), the coefficient for the PUBLIC INFO condition remains negative, statistically significant, and stable in magnitude (ranging between -0.045 and -0.073) relative to the NO INFO benchmark. This stability indicates that the behavioral response is driven by the receipt of the invitation itself, rather than by unobserved heterogeneity in students' social networks or the specific model specification.

Second, the intensity of peer exposure does not alter the decision to participate. In models controlling for network degree (κ_i), the main effect of having more invited peers is statistically negligible (see Column III). Furthermore, the interaction terms between treatment assignment and peer degree (PUBLIC INFO $\times$ κ_i and PRIVATE INFO $\times$ κ_i) are indistinguishable from zero across all network definitions (see Columns IV, VII, and VIII). This implies that a student's sensitivity to the identity disclosure does not vary with the number of invited friends they have, casting doubt on mechanisms based on

social reinforcement or collective action.

Third, there is no evidence of cross-contamination between treatments. The coefficients on cross-treatment exposure (κ_i^{diff}) and their interactions with treatment assignment are consistently small and statistically insignificant (see Columns V and IX). A student assigned to the PUBLIC INFO condition is no more or less likely to participate if they are surrounded by peers in the NO INFO condition. This lack of interference supports the Stable Unit Treatment Value Assumption (SUTVA) and suggests that students did not update their beliefs or behavior based on the differing invitation information received by their classmates.

Finally, Table D-2 reports the analysis disaggregated by academic division. While isolated interaction terms appear significant in specific subsamples, the signs and magnitudes fluctuate unsystematically across divisions. For instance, peer exposure appears to slightly mitigate the negative treatment effect in one division while exacerbating it in another, with no consistent pattern linking tie strength to take-up. This heterogeneity is consistent with statistical noise rather than a structural spillover mechanism.

Table D-2 Peer Spillovers by Academic Division.

OLS estimates of program take-up using the strong-tie network (≥ 10 shared courses). κ_i denotes the number of co-enrolled peers; κ_i^{diff} denotes the number of peers in a different treatment. Columns I–VI correspond to distinct academic divisions. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	I	II	III	IV	V	VI
PUBLIC INFO	−0.116 (0.088)	0.115 (0.178)	−0.296** (0.109)	−0.014 (0.102)	−0.045 (0.108)	0.100 (0.076)
PRIVATE INFO	−0.039 (0.085)	0.037 (0.194)	−0.263* (0.109)	0.015 (0.097)	−0.184+ (0.102)	0.056 (0.074)
κ_i	−0.001** (0.000)	0.002 (0.002)	−0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)
PUBLIC INFO $\times \kappa_i$	0.000 (0.001)	−0.002 (0.003)	0.004* (0.002)	−0.001 (0.002)	−0.000 (0.001)	−0.002* (0.001)
PRIVATE INFO $\times \kappa_i$	−0.000 (0.001)	−0.000 (0.003)	0.003 (0.002)	−0.001 (0.001)	0.002 (0.001)	−0.001 (0.001)
Constant	0.511** (0.064)	0.181 (0.119)	0.394** (0.080)	0.311** (0.074)	0.307** (0.072)	0.156** (0.051)
# Obs.	904	284	453	626	374	633
R^2	0.047	0.008	0.030	0.004	0.013	0.014

Taken together, the evidence in this appendix confirms that the main experimental

findings are not driven by spillovers. The negative effect of public targeting is a direct, individual-level response to the identity signal, unaffected by the density or composition of a student's social network.

D Heterogeneity Analysis

This appendix examines whether the deterrent effect of public targeting is concentrated within specific demographic subgroups or represents a generalized response. I estimate treatment effects separately by academic performance, identity profile complexity, and specific target populations.

D.1 By Performance Group

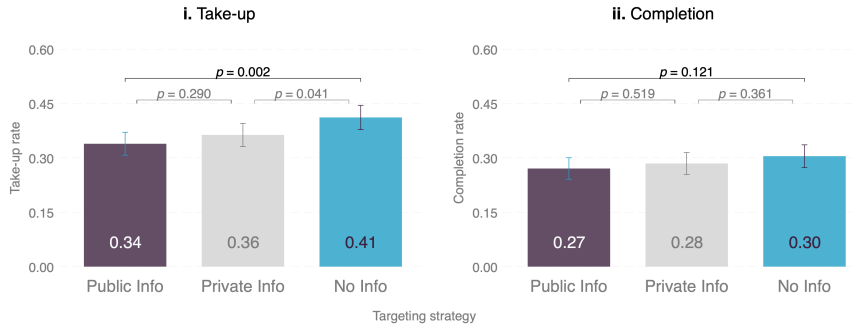
Figure C-1 demonstrates that the response to public targeting is uniform across academic contexts. Despite the difference in framing, where high performers (Wave 1) might view the invitation as a reward and low performers (Wave 2) as remedial support, the PUBLIC INFO condition significantly reduces participation relative to NO INFO in both cohorts. Among high performers, concealing the identity criteria increases take-up by 7 percentage points (a 21% increase); among low performers, it generates a matching 7 percentage point gain (a 44% increase). Completion rates follow an identical pattern, confirming that the behavioral barrier operates independently of the invitation's implied signal regarding student ability (see Appendix B for full regression estimates).

D.2 By Identity Profile

I next test whether the response depends on the intensity of the identity signal. Figure C-2 shows that the negative effect of PUBLIC INFO is robust across profiles: significant reductions in take-up and completion occur for students holding a single targeted identity as well as those holding multiple identities. This consistency applies regardless of profile complexity: the sample includes students with two (40.5%), three (21.5%), or more (3.8%) targeted identities, a distribution that is balanced across treatments in both waves ($p > 0.3$).

A. Information and participation: *High performance*

Take-up and completion for high performers (wave 1)



B. Information and participation: *Low performance*

Take-up and completion for low performers (wave 2)

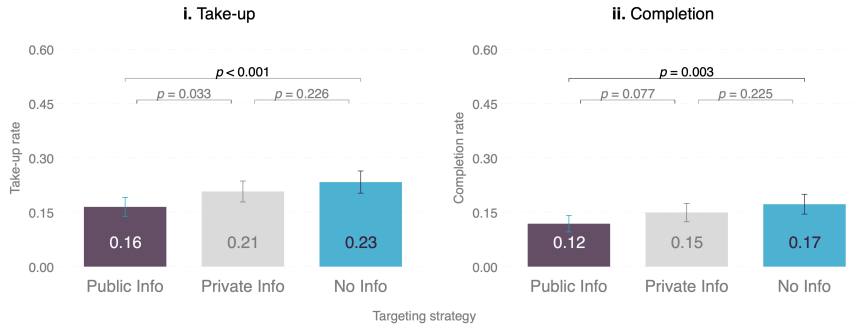


Figure C-1 Participation Rates by Performance Group

Take-up (i) and Completion (ii) rates for High Performance (Panel A) and Low Performance (Panel B) waves. *P*-values from two-sided proportion tests.

Furthermore, Figure C-3 indicates that the result is not driven by any single demographic category. The treatment effect is replicable across the largest subgroups with sufficient statistical power, including female, low-SES, and rural-origin students.

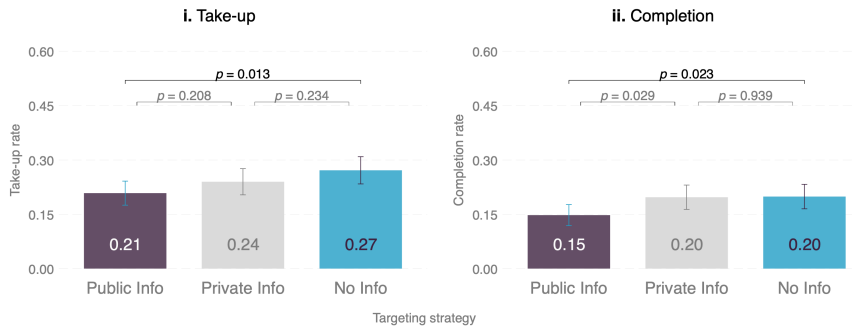
D.3 Completion

The main text established that public targeting depresses program participation at the extensive margin (take-up) but not the *conditional* intensive margin (completion conditional on take-up). Figure C-4 confirms that this pattern holds strictly within both performance waves.

Among high performers, the average number of completed steps drops from 3.47 in NO INFO to 2.99 in PUBLIC INFO ($p = 0.022$); among low performers, it drops from 1.95 to 1.38 ($p = 0.002$). However, conditional on registration (Step 1), these treatment

A. Information and participation: *Single identity*

Take-up and completion for single-identity holders



B. Information and participation: *Multiple identities*

Take-up and completion for multiple-identity holders

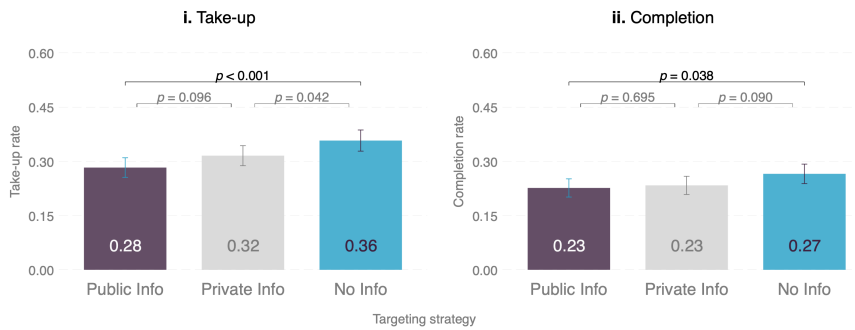


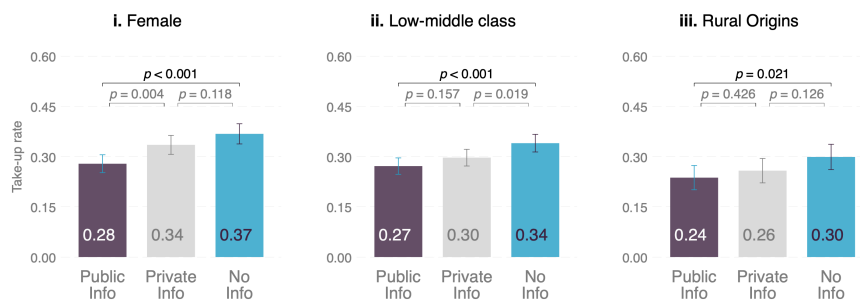
Figure C-2 Participation Rates by Number of Targeted Identities.

Participation rates for students with a single targeted identity (Panel A) versus multiple identities (Panel B). P -values from two-sided proportion tests.

differences vanish in both groups ($p > 0.10$ for all comparisons; see Tables B-2 and B-3 in Appendix B). This disaggregated analysis reinforces the conclusion that the identity friction acts exclusively as a gatekeeping mechanism at entry, preventing students from starting the program rather than causing attrition among those who accept the invitation.

A. Information and take-up: *By specific identities*

Take-up for female, low-middle class and rural students



B. Information and completion: *By specific identities*

Completion for female, low-middle class and rural students

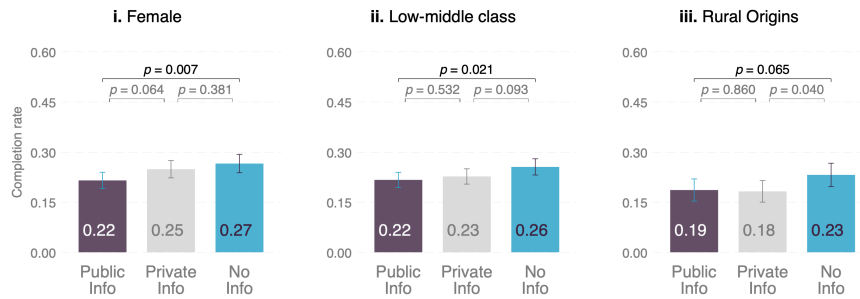
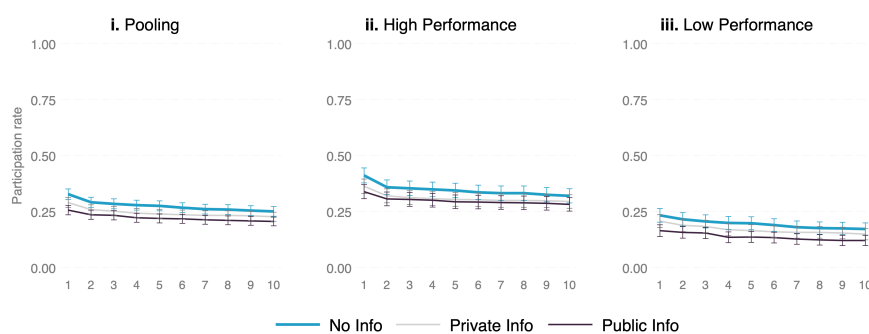


Figure C-3 Participation Rates by Specific Identity Group.

Take-up (Panel A) and Completion (Panel B) rates disaggregated by female, low-SES, and rural status. P -values from two-sided proportion tests.

A. Information and completed steps *unconditional* on take-up



B. Information and completed steps *conditional* on take-up

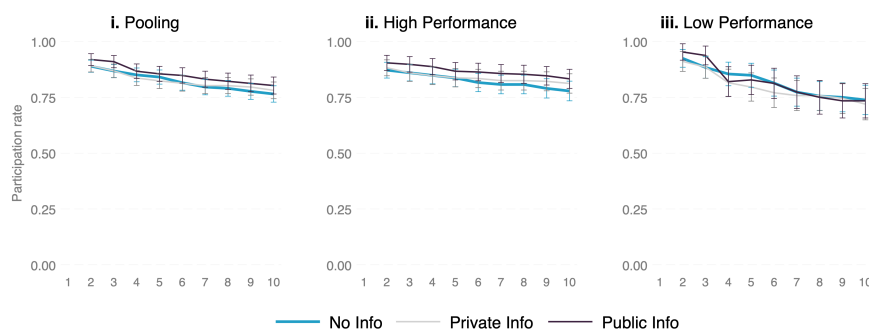


Figure C-4 Number of steps completed in the program.

The figure illustrates how variations in information disclosure impact the rate of completed steps in the program, unconditional on take-up (Panel A) and conditional on it (Panel B).

E Welfare Analysis Details

This appendix quantifies the welfare loss generated by the reduction in take-up under public targeting. I first estimate the causal impact of the training program on academic performance and then calculate the aggregate human capital loss resulting from the deterrent effect of the PUBLIC INFO condition.

E.1 Econometric Strategy

To estimate the returns to program completion, I employ a Difference-in-Differences (DiD) design using administrative panel data. The dataset covers the 4,831 invited students over five academic periods ($N = 21,641$ student-semester observations). The outcome variable is the student's standardized GPA (Z-score), allowing effect sizes to be interpreted in standard deviation (SD) units.

To address selection bias arising from voluntary participation, I estimate the Average Treatment Effect on the Treated (ATT) using a two-way fixed effects model:

$$GPA_{it} = \alpha_i + \delta_t + \beta \cdot (Post_{it} \times Completed_i) + \epsilon_{it} \quad (E-1)$$

where GPA_{it} is the standardized grade point average of student i in semester t . The term α_i represents student fixed effects, which control for time-invariant unobservables such as innate ability. δ_t represents time fixed effects. The coefficient of interest, β , captures the within-student change in performance attributable to completing the program relative to the trajectory of non-participants.

The validity of this design relies on the parallel trends assumption. To validate this assumption and examine the dynamics of the treatment effect, I estimate the following event study specification:

$$GPA_{it} = \alpha_i + \delta_t + \sum_{k=1}^5 \beta_k \cdot (1[Period_{it} = k] \times Completed_i) + \epsilon_{it} \quad (E-2)$$

Periods -1 and 0 are pre-treatment (where Period 0 is the semester of delivery), while Periods 1, 2, and 3 are post-treatment. The coefficients β_k capture the differential GPA

change for treated students in period k .

E.2 Results

Table 2 in the main text reports the static DiD results (Model I), showing that program completion increases GPA by 0.034 SD ($p < 0.001$). Model II reports the dynamic estimation. The pre-treatment coefficients are small and statistically indistinguishable from zero, supporting the parallel trends assumption and indicating that compliers and non-compliers were on similar academic trajectories prior to the intervention.

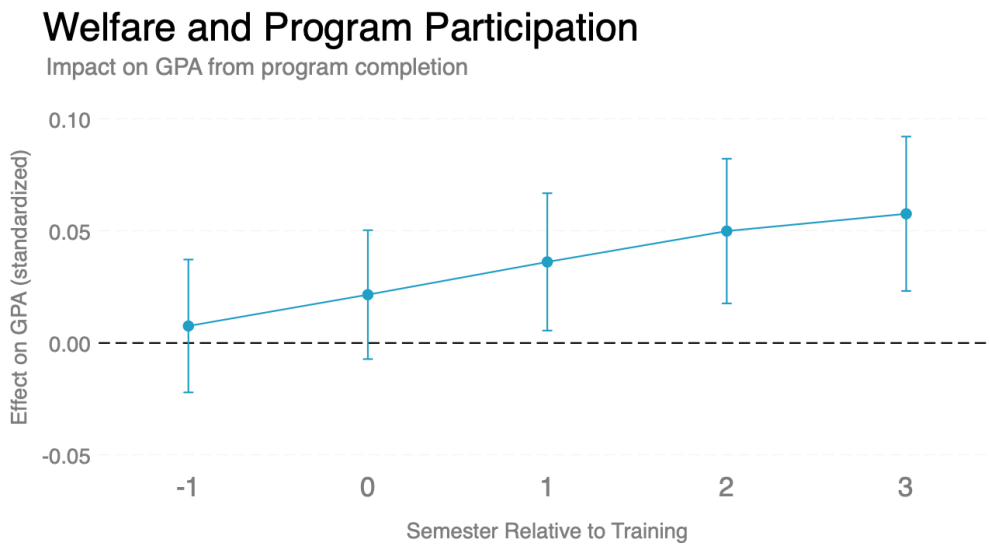


Figure E-1 Impact of Program Completion on GPA (Event Study).

Coefficients from the event study specification (Model II) estimating the effect of program completion on standardized GPA relative to the semester of training ($t = 0$). Error bars represent 95% confidence intervals.

Figure E-1 plots the event-study coefficients. Following the intervention, GPA effects become statistically significant in Period 1 (0.036 SD, $p = 0.020$) and grow over time, reaching 0.050 SD in Period 2 ($p = 0.017$) and 0.058 SD in Period 3 ($p = 0.018$). This pattern suggests that the non-cognitive skills acquired during the training yield cumulative academic benefits.

E.3 Benchmarking Effect Sizes

The estimated impact (0.034 SD on average, growing to 0.058 SD) is economically meaningful when contextualized within the education literature. The program is an

asynchronous, MOOC-style course consisting of nine sessions (approximately 4.5 hours total). Because the marginal cost of treating an additional student is negligible, the intervention is highly scalable.

The observed improvement implies *far transfer*, as students appear to apply the general non-cognitive skills taught in the program (e.g., self-efficacy) across their specific coursework. I benchmark this effect against two common types of interventions. First, *high-touch* interventions such as intensive coaching often yield larger GPA impacts (0.15 to 0.30 SD; e.g., [Bettinger and Baker 2014](#)) but are difficult to scale due to high costs per student. Second, *low-touch* interventions such as informational nudges are inexpensive but rarely improve GPA, often yielding null results (e.g., [Oreopoulos and Petronijevic 2018](#)).

This program occupies a high-value middle ground. It outperforms the nudge baseline, demonstrating that structured content delivers benefits that pure information does not. Achieving an impact of 0.034 to 0.058 SD at near-zero marginal cost represents a substantial Return on Investment (ROI) and offers a scalable policy tool for improving academic outcomes.

E.4 Heterogeneity in Program Impact

To assess whether the costs of non-participation vary across demographic groups, I replicate the event study analysis for the key targeted subpopulations: female students, students from low socio-economic status (SES), and students from rural origins.

Figure [E-2](#) reveals that the program is particularly effective for the primary target groups. Female students (the largest targeted share) and low-SES students exhibit robust positive effects. Results for rural-origin students are estimated with less precision due to smaller sample sizes, though point estimates remain positive. These findings imply that the welfare loss from reduced take-up is most acute for female and low-SES students, as these groups stand to gain the most from the training.

Welfare and Program Participation, by groups

Impact on GPA from program completion

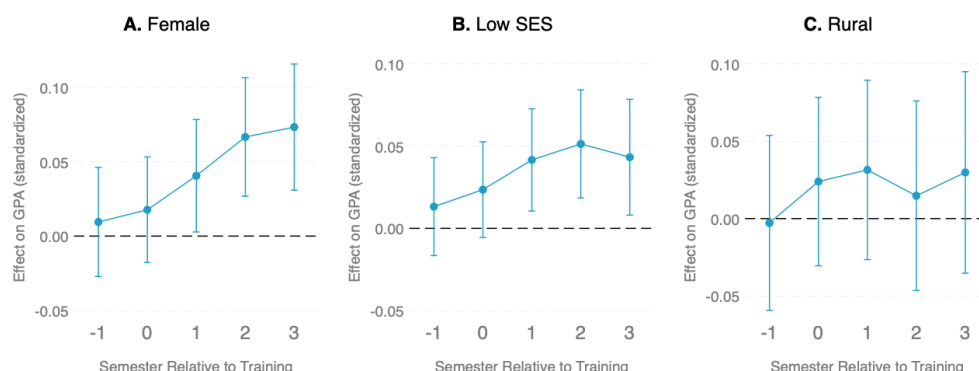


Figure E-2 Heterogeneity in GPA Impacts.

Coefficients from the event study specification estimating the effect of program completion on standardized GPA relative to the semester of training ($t = 0$), separately for Female (A), Low-SES (B), and Rural (C) students. Error bars represent 95% confidence intervals.

E.5 Welfare Cost of Targeting

The validation of the program's efficacy allows me to quantify the welfare cost of the behavioral response to public targeting. The PUBLIC INFO treatment reduced program take-up by 7 percentage points (a 27% decrease) relative to the NO INFO condition. Because completion rates conditional on take-up were similar across treatments, this reduction implies that the PUBLIC INFO framing deterred students who would have otherwise completed and benefited from the program.

Consequently, the targeting strategy actively inhibited human capital accumulation. By adopting the NO INFO strategy, the institution could increase the total human capital generated by the program by 27% without any additional expenditure. The image concerns triggered by public targeting therefore function as a regressive friction: they block access to a welfare-enhancing opportunity for the very groups that the policy intends to support.

F Program Evaluation

At the end of the training, participants were asked to complete an evaluation of the program. 1063 students out of 1066 (99%) who completed the training filled out the

evaluation: 734 in the first wave (High GPA) and 329 in the second wave (Low GPA). Below, I include the main text and items of the questionnaire translated to English (by the author), as the original one was in Spanish. I also report descriptives of the main outcomes and results from a comparative analysis between program waves.

The questionnaire

Please answer to the following questions according to your own experience. There are no right or wrong answers.

1. What is your general evaluation of the program?

[Excellent (5), Good (4), Adequate (3), Deficient (2), Very deficient (1)]

2. The program objectives were clearly stated.

[Strongly agree (5), Agree (4), Neutral (3), Disagree (2), Strongly disagree (1)]

3. The program was well organized.

[Strongly agree (5), Agree (4), Neutral (3), Disagree (2), Strongly disagree (1)]

4. The program was intellectually stimulating.

[Strongly agree (5), Agree (4), Neutral (3), Disagree (2), Strongly disagree (1)]

5. The program helped me improve my thinking.

[Strongly agree (5), Agree (4), Neutral (3), Disagree (2), Strongly disagree (1)]

6. What is your general evaluation of the online sessions?

[Excellent (5), Good (4), Adequate (3), Deficient (2), Very deficient (1)]

7. What is your general evaluation of the instructor?

[Excellent (5), Good (4), Adequate (3), Deficient (2), Very deficient (1)]

8. The instructor encourages learning.

[Strongly agree (5), Agree (4), Neutral (3), Disagree (2), Strongly disagree (1)]

F.1 Evaluation Results

Evaluation of the training program

N= 1063

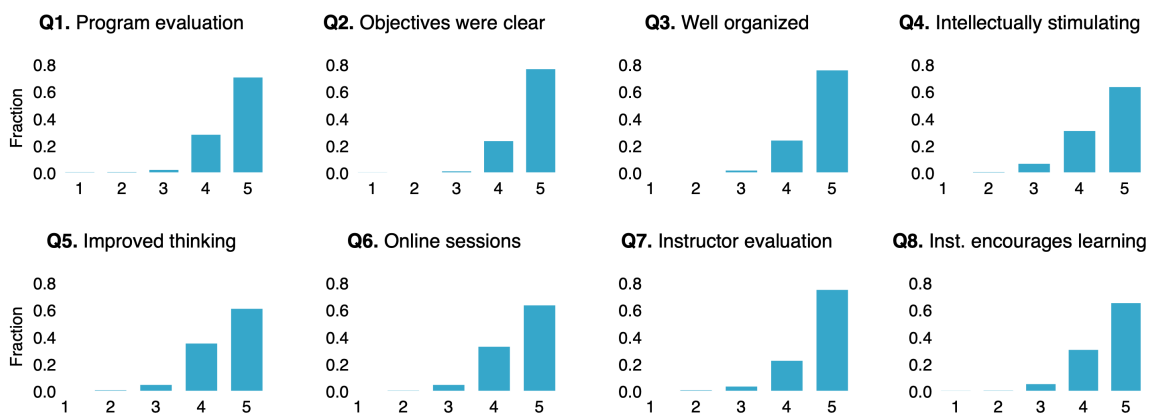


Figure F-1 Program Evaluation Scores.

Distribution of student responses to the post-program evaluation survey ($N = 1,063$).

Figure F-1 reports the outcomes of the program evaluation. Panel Q1 shows that the general program evaluation was high: 97.74% of the responses were either Good (27.75%) or Excellent (69.99%). Panel Q2 shows that 99.16% of responses considered the objectives of the program were clearly explaining: 23.05% Agree and 76.11% Completely agree. Panel Q3 reports that 98.69% of participants Agreed (23.45%) and Completely agreed (75.24%) the program was well organized. Panel Q4 shows that 93.41% considered the program was intellectually stimulating: 30.51% Agreed and 62.90% Completely agreed. Panel Q5 reports that 95.29% of participants believed the program helped them improve their way of thinking: 34.84% agreed and 60.45% completely agreed. Panel Q6 shows that 95.48% of participants consider the online sessions were Good (32.49%) and Excellent (62.99%). Panel Q7 reports that 96.52% participants evaluated the course instructor as Good (22.11%) and Excellent (74.41%). Panel Q8 shows that 94.82% of participants considered the instructor motivated their learning: 30.16% Agree and 64.66%

Completely agree.

The figure and descriptives above pool both waves of the program together. The tables below report regression results for program evaluations by wave, comparing high- and low-performing students across the eight questionnaire items. Consistent with the descriptive evidence, there are no meaningful differences by targeting strategy or wave.

Table F-1 Program Evaluations by Treatment and Wave.

OLS estimates. Dependent variables Q1–Q8 correspond to the survey items detailed in Appendix F. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Public	-0.082 (0.075)	0.037 (0.071)	0.061 (0.064)	-0.021 (0.088)	0.061 (0.085)	-0.044 (0.085)	-0.005 (0.078)	0.004 (0.086)
Private	-0.051 (0.070)	-0.012 (0.058)	-0.006 (0.066)	-0.099 (0.081)	0.058 (0.078)	0.059 (0.074)	0.012 (0.074)	-0.026 (0.081)
High gpa	-0.063 (0.055)	0.011 (0.048)	0.015 (0.055)	-0.039 (0.066)	0.017 (0.069)	-0.019 (0.066)	-0.012 (0.059)	-0.004 (0.067)
Public # High gpa	0.130 (0.089)	0.026 (0.081)	0.009 (0.075)	-0.000 (0.105)	-0.007 (0.101)	0.125 (0.100)	0.046 (0.091)	0.088 (0.101)
Private # High gpa	0.083 (0.086)	-0.009 (0.071)	0.008 (0.079)	0.102 (0.098)	-0.097 (0.094)	-0.040 (0.091)	-0.031 (0.089)	0.066 (0.098)
Constant	4.709*** (0.042)	4.732*** (0.039)	4.709*** (0.046)	4.603*** (0.053)	4.528*** (0.058)	4.571*** (0.056)	4.709*** (0.049)	4.567*** (0.055)
Observations	1063	1063	1062	1062	1062	1062	1063	1061

Table F-2 Program Evaluations with Demographic Controls.

OLS estimates. Dependent variables Q1–Q8 correspond to the survey items detailed in Appendix F. Robust standard errors in parentheses. ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Public	0.013 (0.041)	0.054 (0.035)	0.065* (0.034)	-0.019 (0.048)	0.064 (0.046)	0.045 (0.045)	0.024 (0.040)	0.062 (0.045)
Private	0.012 (0.041)	-0.022 (0.034)	-0.003 (0.037)	-0.029 (0.046)	-0.005 (0.044)	0.030 (0.043)	-0.015 (0.041)	0.012 (0.046)
High gpa	0.002 (0.036)	0.019 (0.032)	0.016 (0.032)	-0.002 (0.042)	-0.008 (0.040)	0.005 (0.040)	-0.008 (0.037)	0.045 (0.041)
Female	-0.052 (0.035)	0.019 (0.032)	0.019 (0.033)	0.034 (0.044)	0.065 (0.042)	0.024 (0.041)	0.104*** (0.039)	0.095** (0.044)
Mid-low class	0.138** (0.063)	-0.032 (0.047)	-0.007 (0.046)	0.036 (0.065)	0.060 (0.067)	0.014 (0.059)	0.072 (0.062)	-0.044 (0.060)
Rural	0.013 (0.037)	-0.040 (0.031)	-0.018 (0.032)	-0.013 (0.042)	0.063 (0.039)	-0.008 (0.039)	-0.001 (0.037)	0.009 (0.040)
First generation	-0.041 (0.042)	0.002 (0.035)	-0.004 (0.036)	-0.103** (0.050)	-0.098** (0.046)	-0.028 (0.045)	0.017 (0.041)	0.005 (0.044)
Ethnic minority	-0.185 (0.154)	0.165*** (0.062)	-0.088 (0.094)	0.008 (0.100)	0.214*** (0.080)	0.026 (0.109)	0.063 (0.082)	0.134 (0.092)
Constant	4.583*** (0.075)	4.751*** (0.058)	4.711*** (0.061)	4.544*** (0.083)	4.434*** (0.086)	4.533*** (0.078)	4.565*** (0.076)	4.502*** (0.081)
Observations	1063	1063	1062	1062	1062	1062	1063	1061

G Mechanism Experiments

In this appendix, I report three lab-in-the-field experiments conducted to further explore the mechanisms driving the effect of information disclosure on program take-up. Figure G-1, illustrates the timeline of all the experiments reported in this paper.

Across the three lab-in-the-field experiments, the main setup is the same. So, I first describe the common parts of all studies, and then expand on each case separately. Below, I include the main text and items of the questionnaire translated to English (by the author), as the original experiments were in Spanish.

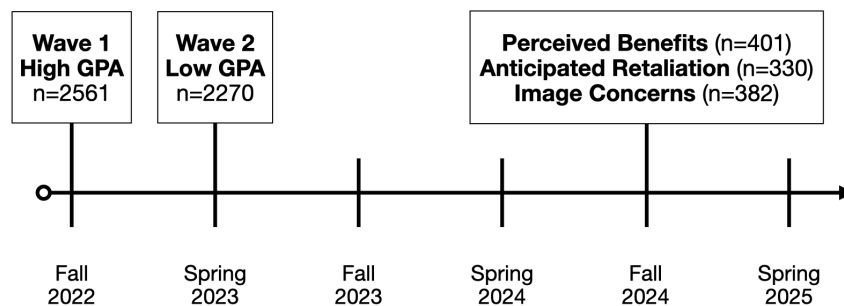


Figure G-1 Timeline of Experiments.

Chronology of the field intervention waves and the complementary mechanism experiments.

Part 1 is the same across experiments. In this part, I collect demographics: gender, socio-economic status and GPA. I use gender and socio-economic status to block randomize students into treatments in Part 2. There are two conditions: NO INFO vs. INFO. The main difference is that the invitation to the program discloses that selection is based on group identities (see below). I also use the GPA to personalize the invitation, the same way it was done for the field experiment.

G.1 Common Experimental Setup

PART 1

The first questions are about you.

- What is your gender? [*male-female*]
- Which socio-economic strata does your family belong to? [*stratum 1 to stratum 6*]
- What is your cumulative grade point average? (if you don't remember exactly, give us your best guess)

PART 2

To answer the following question(s), you will need to watch a short video (below 2 minutes).

This video describes the international training program that we want to offer some students at [local university], in alliance with [international university].

Attention: *The video has audio. Please turn up the volume or use headphones.*

[page break]

Please watch the following video carefully.

If you are using a cellphone, turn it horizontally for better viewing.

https://youtu.be/bKeH9kUoSqo?si=P2atu_UsfABFR3tB

[page break]

If you were selected to participate in the program, you would receive the following invitation:

- NO INFO: You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA].
- INFO: You have been chosen among all students at the university because you can benefit from this program, as your cumulative GPA is [Student's GPA], and because you fulfill one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a first-generation student (neither of your parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

G.2 Experiment 1: Perceived Benefits

To assess the impact of information disclosure on the anticipated benefits from the program (and test Hypothesis 2), I conducted an online experiment with 401 students from the local university. 214 were in INFO and 187 in NO INFO. I randomized participants into treatments through block randomization using female and socio-economic status (right after Part 1). The randomization is balanced for socio-economic status and GPA (note, I do not use GPA to randomize but control for it being balanced between treatments), while for females there is more representation in the NO INFO condition (see Table G-1 below).

Table G-1 Balance in Perceived Benefits Experiment.

Mean comparisons of randomization variables between NO INFO and INFO conditions. Column III reports p -values from t -tests of equality.

	NO INFO I	INFO II	p-value III
Female	0.67 (0.03)	0.57 (0.03)	0.042
Low SES	0.89 (0.02)	0.90 (0.02)	0.908
GPA	4.21 (0.02)	4.16 (0.02)	0.131
Observations	187	214	

At the end of Part 2, I elicited the personal valuation. Participants who completed this section could earn a fixed bonus. Part 3 is a belief elicitation stage and depending on accuracy, participants could earn an additional bonus, see the questionnaire below.

The questionnaire

The following section was displayed at the end of Part 2.

- How much value, utility or benefit would the course have for you?

Answer using a scale from 0 to 10, where 0 means “Very low value, utility or benefit” and 10 means “Very high value, utility or benefit” [0-10]

[page break]

You have completed your participation in this survey and will be included in the lottery to get one of the monetary bonuses.

You can end your participation here. But, you also have the chance to participate in an additional lottery for another bonus of 50 thousand pesos, if you answer some multiple-choice questions about your beliefs regarding the course.

Among those who answer the 9 additional questions, we will randomly allocate ten monetary bonuses of 50 thousand pesos. So you can participate independently in two lotteries **and may win up to 100 thousand pesos**.

if you want to participate click on **Continue**. If you want to conclude your participation here, choose **End**.

[page break]

PART 3

More than one thousand students at [local university] have completed the training program, and to all of them we have asked them to indicate how they rate the program, using the following options: Excellent, Good, Adequate, Deficient, Very Deficient.

We want to know your opinion about the evaluation that 9 groups of students gave to the program. For each of these groups, please indicate which of the options you believe was chosen by most people.

If you are chosen for the lottery, the computer will randomly choose one of your 9 answers. If the option you indicate is the one that most participants in the course used for their evaluation, **you will win one of the 10 additional bonuses of 50 thousand pesos**.

[**Note:** the items were displayed in random order.]

- Which do you think is the program evaluation chosen by most **male** students?
- Which do you think is the program evaluation chosen by most **female** students?
- Which do you think is the program evaluation chosen by most **low class (strata 1 and 2)** students?

- Which do you think is the program evaluation chosen by most **middle class (strata 3 and 4)** students?
- Which do you think is the program evaluation chosen by most **high class (strata 5 and 6)** students?
- Which do you think is the program evaluation chosen by most **first generation** (neither of their parents holds a college degree) students?
- Which do you think is the program evaluation chosen by most **continuous generation** (at least one of their parents holds a college degree) students?
- Which do you think is the program evaluation chosen by most students **from rural areas**?
- Which do you think is the program evaluation chosen by most students **from urban areas**?

Additional results on perceived benefits

In Part 3 of the experiment, I elicited the beliefs students have about the most frequent course evaluation (i.e., Question 1 from the evaluation questionnaire, as reported in Appendix F) provided by each social group. The main outcome from the comparison indicates that beliefs about the course evaluation are very high (4.2 out of 5) in the experiment, although lower than in the actual course evaluation: 4.6 ($p < 0.001$). Figure G-2 reports the average evaluation of the program for each social category, contrasting the beliefs reported in the experiment and the behavior in the actual course evaluation.

G.3 Experiment 2: Endorser Retaliation

To assess the impact of information disclosure on the anticipated retaliation from the endorser (and test for Hypothesis 3), I conducted an online experiment with 330 students from the local university. 158 were in INFO and 172 in NO INFO. As in the other experiments, I randomized participants into treatments through block randomization using female and socio-economic status, after Part 1. As in the other experiments, I

Program evaluation: Beliefs vs. Behavior



Figure G-2 Beliefs vs. Behavior in Course Evaluation.

Comparison of elicited beliefs regarding peer course evaluations (dark bars) versus actual evaluations submitted by program participants (light bars).

check for female, socio-economic status and GPA, all of which are balanced between treatments (see Table G-2 below).

Table G-2 Balance in Anticipated Retaliation Experiment.

Mean comparisons of randomization variables between NO INFO and INFO conditions. Column III reports *p*-values from *t*-tests of equality.

	NO INFO	INFO	p-value
	I	II	III
Female	0.63 (0.03)	0.63 (0.03)	0.987
Low SES	0.93 (0.02)	0.92 (0.02)	0.829
GPA	4.22 (0.02)	4.17 (0.03)	0.192
Observations	172	158	

In Part 2, this experiment included the pre-defined endorsement message that students were required to send the faculty endorser to request their support:

- NO INFO: I, [Professor's name] endorse student [Your Name] to take part in the training program "How to change: scientific tools to achieve the goals in your

personal and professional life”, because he/she can benefit from this program, as his/her cumulative GPA is [Student’s GPA].

- INFO: I, [Professor’s name] endorse student [Your Name] to take part in the training program “How to change: scientific tools to achieve the goals in your personal and professional life”, because he/she can benefit from this program, as his/her cumulative GPA is [Student’s GPA], and because he/she fulfills at least one of the following requirements: being a woman, being of low socio-economic status, belonging to an ethnic minority (indigenous or afro-descendant), being a first-generation student (neither of his/her parents has a college degree), or coming from a rural area (or not coming from any of the main cities in the country).

Then, in Part 3, I elicited information and beliefs about the endorser. I include the questionnaire below.

The questionnaire

- If you had to request an endorsement from a faculty member to attend the course (asking them to send an email with the text displayed above), who would you ask for it? Please write the name and program of the faculty member you would reach out to (we will not contact him/her).
- If you ask [Professor’s name] for the endorsement, what is the probability that he/she will help you and send the endorsement email? (choose a value between 0 “He/she would not send the endorsement email” and 100 “He/she would certainly send the endorsement email”).
- If you were offered a slot in the international training program, do you believe [Professor’s name] would judge you negatively for it? [Yes - No]
- If you were offered a slot in the international training program, do you believe [Professor’s name] would treat you negatively in the future because you requested the endorsement? [Yes - No]

G.4 Experiment 3: Image Concerns

To further evaluate whether there are costs associated with image concerns that may be triggered through the disclosure of information about selection being identity-based (and further test between Hypotheses H1 and H0), I ran an online experiment with 382 students from the local university. 193 were in INFO and 189 in NO INFO. As in the other experiments, I randomized participants into treatments through block randomization by gender and socio-economic status, after Part 1. The randomization is balanced across female, socio-economic status and GPA (see Table G-3 below).

Table G-3 Balance in Image Concerns Experiment.

Mean comparisons of randomization variables between NO INFO and INFO conditions. Column III reports p -values from t -tests of equality.

	NO INFO I	INFO II	p-value III
Female	0.66 (0.03)	0.68 (0.03)	0.719
Low SES	0.85 (0.02)	0.84 (0.02)	0.954
GPA	4.12 (0.02)	4.10 (0.02)	0.599
Observations	189	193	

This experiment only included the first part of the invitation, without any reference to the endorsement message.

The questionnaire

The following is displayed at the end of Part 2. The order of each block of 3-questions was randomized, as well as the three items within each block.

This is the last part of this survey. Take into account the invitation and answer to 9 questions about the people invited to the program.

Answer to this block of three questions: In your opinion, after receiving an invitation like the one we showed you above...

- **Female** students feel that they receive a placement in the program because of *pity/obligation*. [Never, Rarely, Sometimes, Often, Always]

- **Rural** students feel that they receive a placement in the program because of *pity/obligation*. [Never, Rarely, Sometimes, Often, Always]
- **Low SES** students feel that they receive a placement in the program because of *pity/obligation*. [Never, Rarely, Sometimes, Often, Always]

Answer to this block of three questions: In your opinion, after receiving an invitation like the one we showed you above...

- **Female** students feel *attacked/offended* for receiving a placement in the program. [Never, Rarely, Sometimes, Often, Always]
- **Rural** students feel *attacked/offended* for receiving a placement in the program. [Never, Rarely, Sometimes, Often, Always]
- **Low SES** students feel *attacked/offended* for receiving a placement in the program. [Never, Rarely, Sometimes, Often, Always]

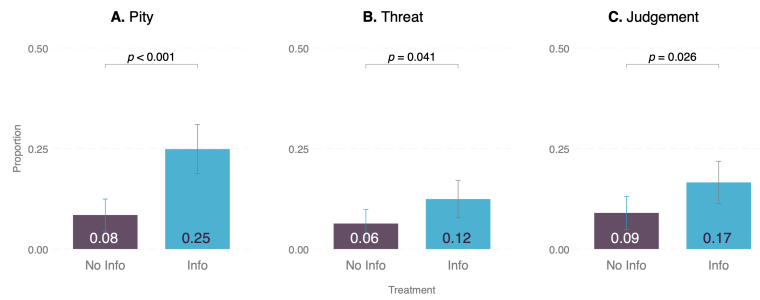
Answer to this block of three questions: In your opinion, after receiving an invitation like the one we showed you above...

- **Female** students feel *judged/stigmatized* by others if they accept their placement in the program. [Never, Rarely, Sometimes, Often, Always]
- **Rural** students feel *judged/stigmatized* by others if they accept their placement in the program. [Never, Rarely, Sometimes, Often, Always]
- **Low SES** students feel *judged/stigmatized* by others if they accept their placement in the program. [Never, Rarely, Sometimes, Often, Always]

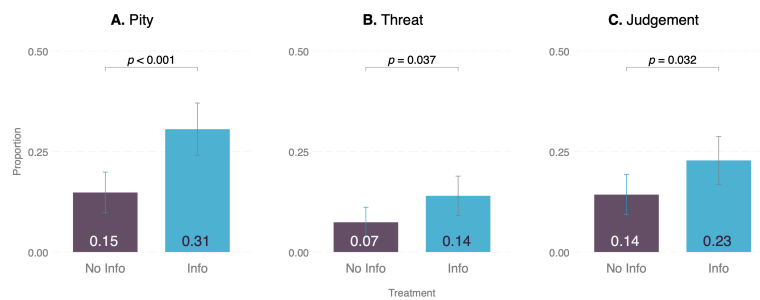
Additional results

In the main text, I report aggregate measures of the image concerns reported in the experiment, pooling together the beliefs for female, rural and low socio-economic status students. In Figure G-3, I illustrate the outcomes separately for each social group. The results across groups are consistent: individuals believe that information about selection being identity-based can trigger image concerns. This is observed for female students, low socio-economic status students and rural students.

Information and image concerns: Female



Information and image concerns: Low-Middle class



Information and image concerns: Rural

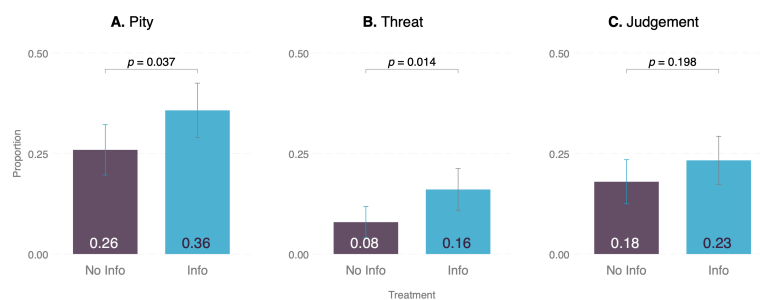


Figure G-3 Image Concerns by Social Group.

Elicited beliefs regarding feelings of pity, threat, and judgment for Female (Top), Low-SES (Middle), and Rural (Bottom) students. P -values from two-sided t -tests.

H Pre-registration and Ethics

The field experiment was conducted in partnership with the local university and implemented in two waves: High performers in the Fall of 2022 and Low performers in the Spring of 2023. This section details the pre-registration process, the administrative motivation for the study timeline, and the additional analyses conducted beyond the original protocol.

H.1 Pre-registration and Study Timeline

The staggered implementation of the experiment was driven by administrative requirements from the local partner university. To ensure the success of the new alliance with the international partner, the local university requested that the program be offered first to high-performing students (Wave 1), anticipating that this group would show higher initial engagement and participation. Following the successful implementation of the first wave, the local partner authorized the expansion of the program to low-performing students (Wave 2).

The study was formally pre-registered prior to the launch of the second wave (Spring 2023). The timing of the pre-registration reflects the administrative process required to finalize the data-sharing agreement for scientific publication. Crucially, the experimental design, intervention materials, randomization blocks, and data collection procedures were identical across both waves. Because the protocol remained unchanged, I pool data from both waves for the main analysis to maximize statistical power, while transparently noting the administrative sequencing.

The data collected in this field experiment are used exclusively for this study. No other interventions were embedded in this project, and no other papers have been or will be generated from this dataset.

H.2 Outcome Measures and Multiple Testing

The primary outcome measure specified in the pre-registration was program take-up. I include program completion as a secondary outcome to assess the quality of the intervention and to isolate the specific margin where the treatment effect operates. The high overall completion rates among participants suggest high program satisfaction, consistent with course evaluations. Furthermore, the analysis reveals that completion rates conditional on take-up do not vary by treatment; this confirms that the behavioral friction introduced by the targeting strategy is concentrated entirely at the extensive margin (take-up).

Because the analysis involves three treatments (PUBLIC INFO, PRIVATE INFO and NO INFO) and two related outcomes (take-up and completion), I address potential concerns

regarding multiple hypothesis testing (MHT). I apply the Holm-Bonferroni correction (Holm 1979), a step-down procedure that controls the family-wise error rate (FWER) while being uniformly more powerful than the classical Bonferroni method.

Table H-1 reports the adjusted p -values for the main treatment effects relative to the NO INFO condition.

Outcome	Treatment	Coefficient	Robust SE	Raw p	Holm p
Take-up	Public	-0.071	0.016	0.000	0.000
Take-up	Private	-0.037	0.016	0.022	0.022
Completion	Public	-0.044	0.015	0.003	0.006
Completion	Private	-0.021	0.015	0.156	0.156

Holm-Bonferroni adjusted p -values for the main treatment effects on take-up and completion.

The main conclusions remain robust to these adjustments. For take-up, both treatments show statistically significant reductions relative to NO INFO, with PUBLIC INFO remaining highly significant (Holm-adjusted $p < 0.001$). For completion, the negative effect of the PUBLIC INFO treatment retains statistical significance (Holm-adjusted $p = 0.006$), confirming that the results are not driven by multiple comparisons.

H.3 Analysis Strategy

The pre-registration specified that the data analysis would be conducted using OLS regressions. Consistent with this, the full regression analysis controlling for covariates is reported in Appendix B. In the main text, I report comparisons using proportion tests to provide a transparent description of the binary outcomes (take-up and completion). The results from the proportion tests and the pre-registered linear probability models are consistent.

In addition to the pre-registered outcomes, I conduct two supplementary analyses to validate the identification strategy and quantify welfare effects. These analyses rely on administrative records that were not part of the original experimental data collection but were matched ex-post:

1. **Peer Spillovers (Appendix C):** To test for potential contamination, I reconstruct

the student social network using course co-enrollment data from the 6 academic periods surrounding the experiment (the 5 semesters prior to the intervention plus the semester of the intervention).

2. **Welfare Analysis (Appendix E):** To estimate the impact of the program on human capital, I use a difference-in-differences design on student GPA. This panel covers 6 academic periods: the 2 semesters prior to the course, the semester of the course, and the 3 subsequent semesters.

These auxiliary analyses are used to rule out SUTVA violations and to benchmark the cost of the behavioral response, respectively.

H.4 Ethics Statement

All studies reported in this paper were approved by the Ethics Review Board at Universidad Autónoma de Bucaramanga, the local partner university. The intervention was implemented as a standard educational opportunity offered by the Office of International Relations of the partner university, and consistent with the nature of a natural field experiment, students were not explicitly informed at the moment of the invitation that they were part of a study. However, the study operates under the university's general consent framework: at the beginning of their studies, all students sign a consent form acknowledging that they may participate in research projects and that their administrative data may be used and shared for research purposes. Students retain the right to rescind this consent at any time. At the time of the study, none of the invited students had exercised this option.